MOJJ ENGINEERING SYSTEMS LTD.

PROCESS DESIGNERS AND PROJECT ENGINEERS
81-B/15 MIDC BHOSARI, PUNE-411 026,INDIA

PROJECT NO	MSD-697
PROJECT NAME	SPRAY DRYER
PANEL	PLC PANEL
END USER	M/S STRING BIO PVT. LTD.

MANUFACTURER	MOJJ ENGINEERING SYSTEMS LTD.
YEAR OF MFG	YEAR 2024-25
DRAWING REVISION	AS BUILT

Edit date 28-02-2025 CREATOR SYSTEM SAKSHI.BHADAURIYA

TOTAL NO. OF PAGES 43

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PROJECT:- MSD-697

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GENERAL SPECIFICATIONS

A. REFERENCES

01	CUSTOMER NAME	:	MOJJ ENGINEERING SYSTEMS. LTD
02	PROJECT	:	PLC PANEL

B. SYSTEM PARAMETERS

01	RATED INCOMING	:	230VAC,16A 1PH. 2WIRE
02	RATED CONTROL	:	230VAC
03	VOLTAGE VARIATION, %	:	± 10%
04	TYPE OF SYSTEM	:	PLC PANEL
05	RATED FREQUENCY, Hz.	:	50 Hz ± 5%
06	AMBIENT TEMP.	:	40°C

C. CONSTRUCTION

01	MAKE OF PANEL	:	RITTAL-AISI 304 TS SERIES		
02	DIMENSIONS IN MM.	:	1800+100(H)X1600(W)X500(D) - FOR PLC CONTROL PANEL		
03	MATERIAL	:	ENCLOSURE FRAME	:	STAINLESS STEEL 1.4301 (AISI 304) 1.5MM
		:	ROOF	:	STAINLESS STEEL 1.4301 (AISI 304) 1.5MM EXTERIOR BRUSHED
		:	DOOR	:	STAINLESS STEEL 1.4301 (AISI 304) 2.0MM EXTERIOR BRUSHED
		:	REAR PANEL	:	STAINLESS STEEL 1.4301 (AISI 304) 1.5MM EXTERIOR BRUSHED
		:	GLAND PLATE	:	STAINLESS STEEL 1.4301 (AISI 304) 1.5MM EXTERIOR BRUSHED
		:	MOUNTING PLATE	:	SHEET STEEL, 3.0MM
04	SURFACE FINISH	:	ENCLOSURE FRAME & GLAND PLATES	:	UNCOATED DOOR, ROOF & REAR PANEL -EXTERIOR BRUSHED, GRAIN SIZE 400
		:	MOUNTING PLATE	:	ZINC PLATED
05	CABLE ENTRY	:	FROM TOP SIDE		
06	LIFTING EYEBOLTS	:	PROVIDED AT TOP		
07	REAR ACCESS	:	NO		
08	DOORS	:	FRONT DOOR OPENING		
09	DEGREE OF PROTECTION	:	IP-52		

D. EQUIPMENT

01	CONTROL/PROTECTION	:	MCCB'S,MCB'S USED

02	LABEL STICKERS	:	ALL LEGEND INSIDE PANEL WILL BE YELLOW STICKERS. YELLOW BACKGROUND WITH BLACK PRINTED LETTERS.
		:	ALL LEGEND OUTSIDE PANEL WILL BE ALLUMINIUM LABELS. BLACK BACKGROUND WITH WHITE PRINTED LETTERS.

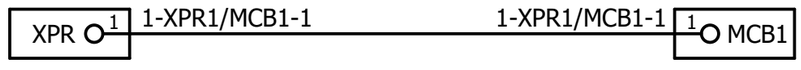
03 WIRING COLOUR CODING

230V AC, 1Ph INCOMING	:	AS PER RATING / RED & BLACK
230V AC, 1Ph (CONTROL WIRING)	:	2.5 SQ.MM RED/BLACK
POWER EARTH	:	2.5 SQ.MM YELLOW/GREEN
L-24VDC	:	0.5 TO 2.5 SQ.MM ORANGE
M-24VDC	:	0.5 TO 2.5 SQ.MM WHITE
DIGITAL INPUT	:	0.5 TO 2.5 SQ.MM BLUE
DIGITAL OUTPUT	:	0.5 TO 2.5 SQ.MM YELLOW
RELAY TO TERMINAL	:	1.5 SQ.MM POTENTIAL FREE(GREY)
ANALOG INPUT	:	0.5 SQ.MM SHIELDED
ANALOG OUTPUT	:	0.5 SQ.MM SHIELDED

05	BUSBAR MOC	:	ALUMINIUM		
	SIZES	:	EARTHING BUSBAR	:	1R X 25X 5MM
		:	INTRUMENT EARTH	:	1R X 25X 5MM
05	BUSBAR MOC	:	COPPER		
	SIZES	:	INTRUMENT EARTH	:	1R X 20X 5MM

06 TERMINAL ARRANGEMENTS

POWER & CONTROL TERMINALS	:	ELMEX
FERRULES	:	CROSS FERRULING
FERRULING TYPE	:	BLACK LETTER ON WHITE SLEEVE (PRINTED FERRULE)
EXAMPLE	:	PI NO.-DESIGNATION/DESIGNATION- PIN NO.



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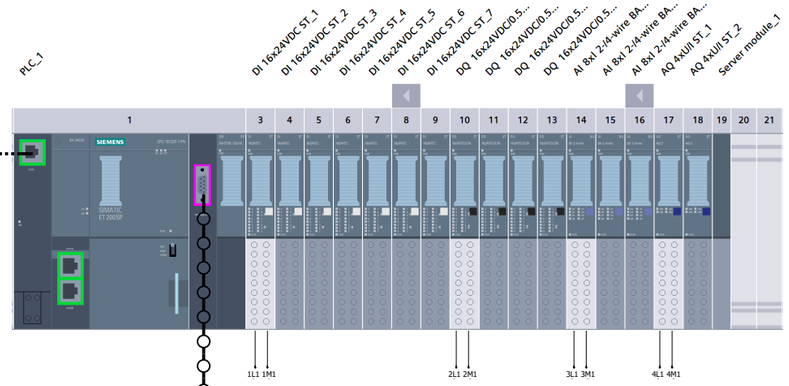
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#PLC SYSTEMS ARCHITECTURE



PLC I/O COUNT		
1	DIGITAL INPUT	112
2	DIGITAL OUTPUT	64
3	ANALOG INPUT	24
4	ANALOG OUTPUT	8

THIRD PARTY COMMUNICATION (PROFIBUS)

ETHERNET SWITCH



CONTROL ROOM

WIN CC V8.0 RT 2048 (2048 POWER TAGS)



SIMATIC HMI MTP1200 UNIFIED COMFORT
(6AV21283MB060AX1)



PROFINET COMMUNICATION

PLC HARDWARE DETAILS

SR.NO.	HARDWARE DETAILS	MLFB NO.	QTY.(NO'S)
01	CPU-1512SP-1 PN FOR ET 200 SP	6ES75121DK010AB0	01
02	PROFIBUS CARD	6ES75455DA000AB0	01
03	SIMATIC ET 200SP, Digital input module, DI 16x 24V DC	6ES71316BH010BA0	07
04	SIMATIC ET 200SP, Digital output module, DQ 16x 24V	6ES71326BH010BA0	04
05	SIMATIC ET 200SP, analog input module, AI 8XU/I 2/4-wire basic	6ES71346GF000AA1	03
06	SIMATIC ET 200SP, Analog output module, AQ 4XU/I	6ES71356HD000BA1	02
07	Unmanaged Industrial Ethernet Switch	64K50080BA20YAB2	01
08	SIMATIC ET 200SP, BaseUnit BU15-P16+A0+2D, BU type A0, push-in terminals	6ES71936BP000DA0	04
09	SIMATIC ET 200SP, BaseUnit BU15-P16+A0+2B, BU type A0, Push-in terminals	6ES71936BP000BA0	12
10	SIMATIC S7 MEMORY CARD 12 MB FOR PLC	6ES7954-8LE03-0AA0	01
11	SIMATIC S7 MEMORY CARD 2GB FOR HMI	6AV21818XP000AX0	01
12	SIMATIC HMI MTP1200 UNIFIED COMFORT	6AV21283MB060AX1	01

PC DETAILS

INTEL CORE I5 PROCESSOR 16-GB RAM 1TB SSD 1 USB OPTICAL SCROLL MOUSE 1 USB KEYBOARD LAN PORT 24" MONITOR MS WINDOWS 11 PROF. MS OFFICE LICENCE CONSIDERED 1YR. ANTIVIRUS	MAKE-DELL

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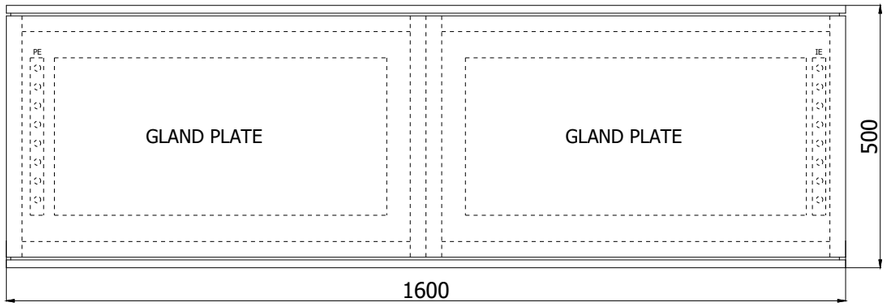
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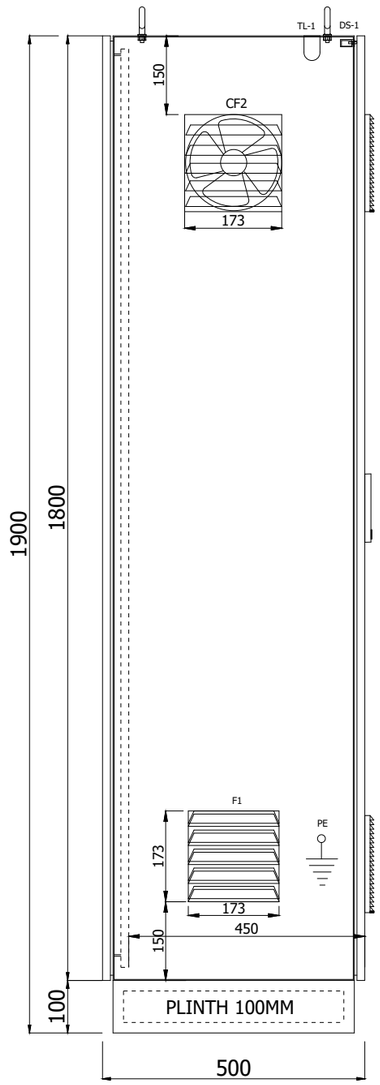
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EXTERNAL GENERAL ARRANGEMENT DRAWING FRONT SIDE

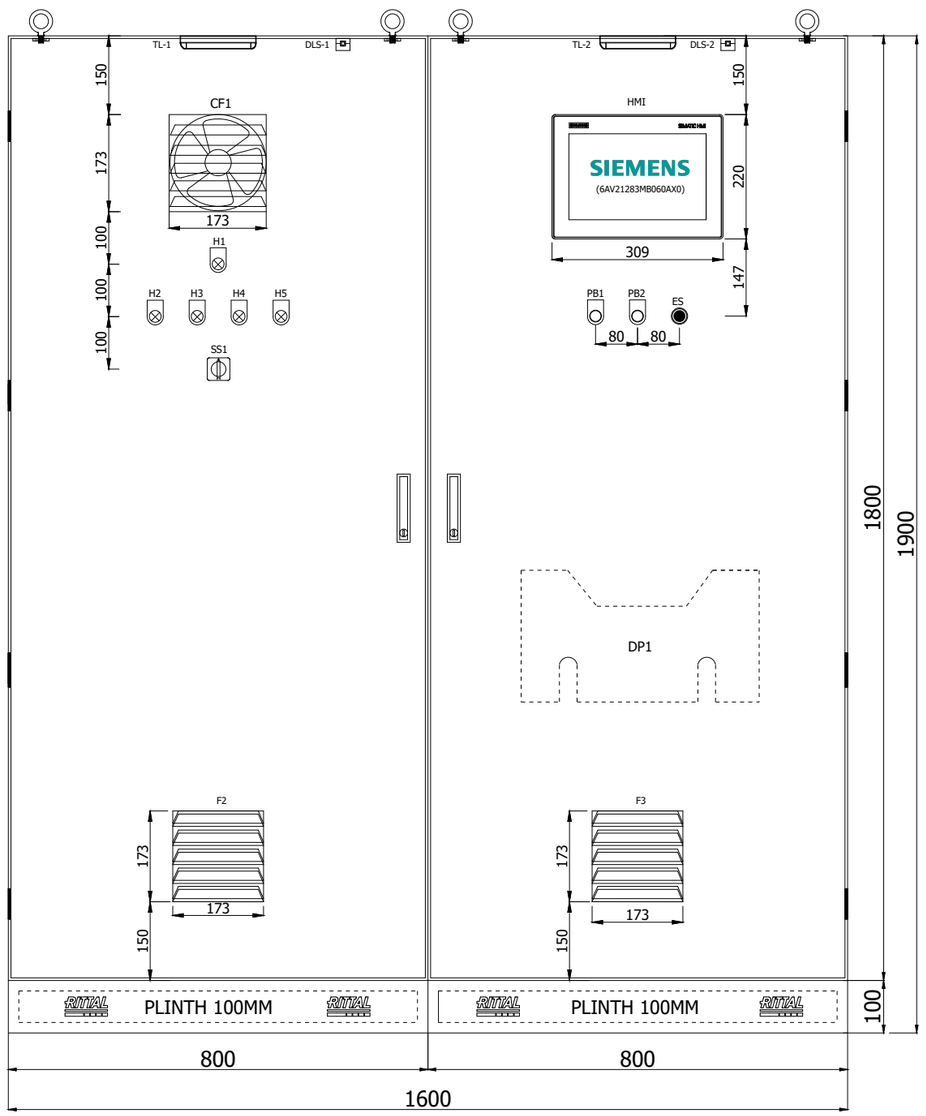
BOTTOM VIEW



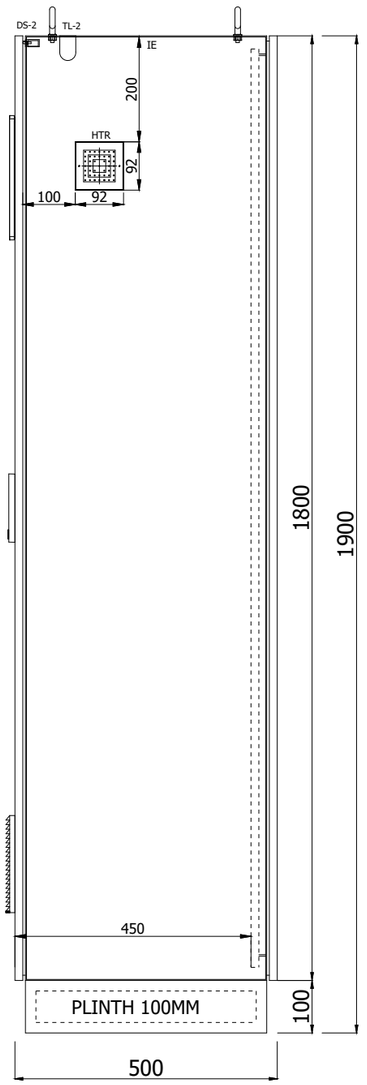
LHS VIEW



FRONT VIEW



RHS VIEW



LEGEND DETAILS:-

TL 1-2	:	PANEL TUBE LIGHT
DL 1-2	:	DOOR LIMIT SWITCH
CF 1-2	:	6"COOLING FAN WITH FILTER
F1-3	:	6" AIR FILTER
H1	:	230VAC ON IND. LAMP (RED)
H2	:	SMPS1, 24VDC ON INDICATION LAMP (GREEN)
H3	:	SMPS2, 24VDC ON INDICATION LAMP (GREEN)
H4	:	SMPS3, 24VDC ON INDICATION LAMP (GREEN)
H5	:	SMPS4, 24VDC ON INDICATION LAMP (GREEN)
SS1	:	MAINS ON/OFF SELECTOR SWITCH
PB1	:	HOOTER ACK. PUSH BUTTON (YELLOW)
PB2	:	FAULT RESET PUSH BUTTON (BLACK)
PB3	:	EMERGENCY STOP MUSHROOM HEAD PUSH BUTTON
IE	:	INSTRUMENT EARTHING BUSBAR
PE	:	POWER EARTHING BUSBAR
HTR	:	HOOTER 24VDC
DP1	:	DRAWING POCKET
IE	:	INSTRUMENT EARTHING

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TITLE:- EXTERNAL GENERAL ARRANGEMENT DRAWING

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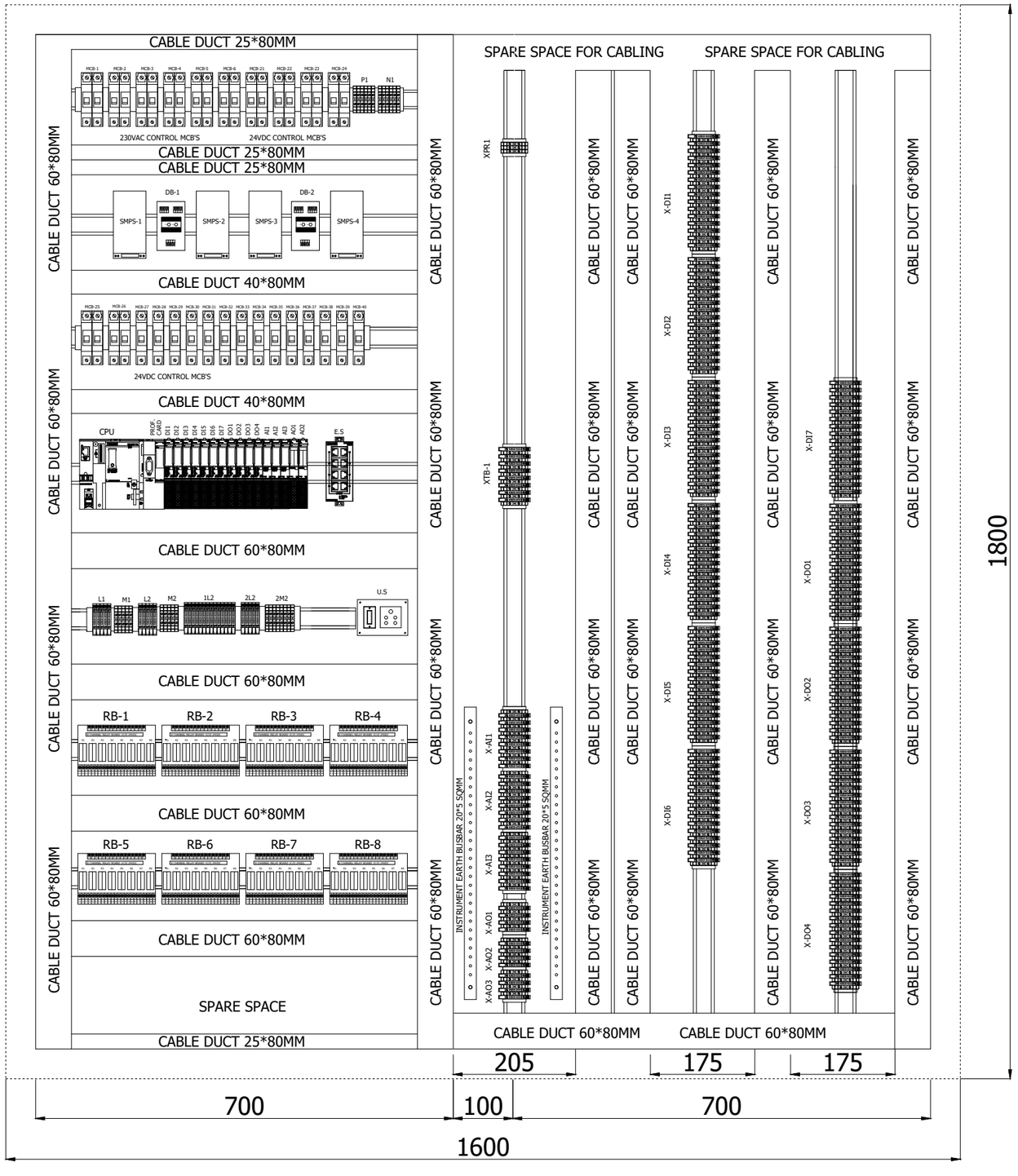
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INTERNAL GENERAL ARRANGEMENT DRAWING

MOUNTING VIEW



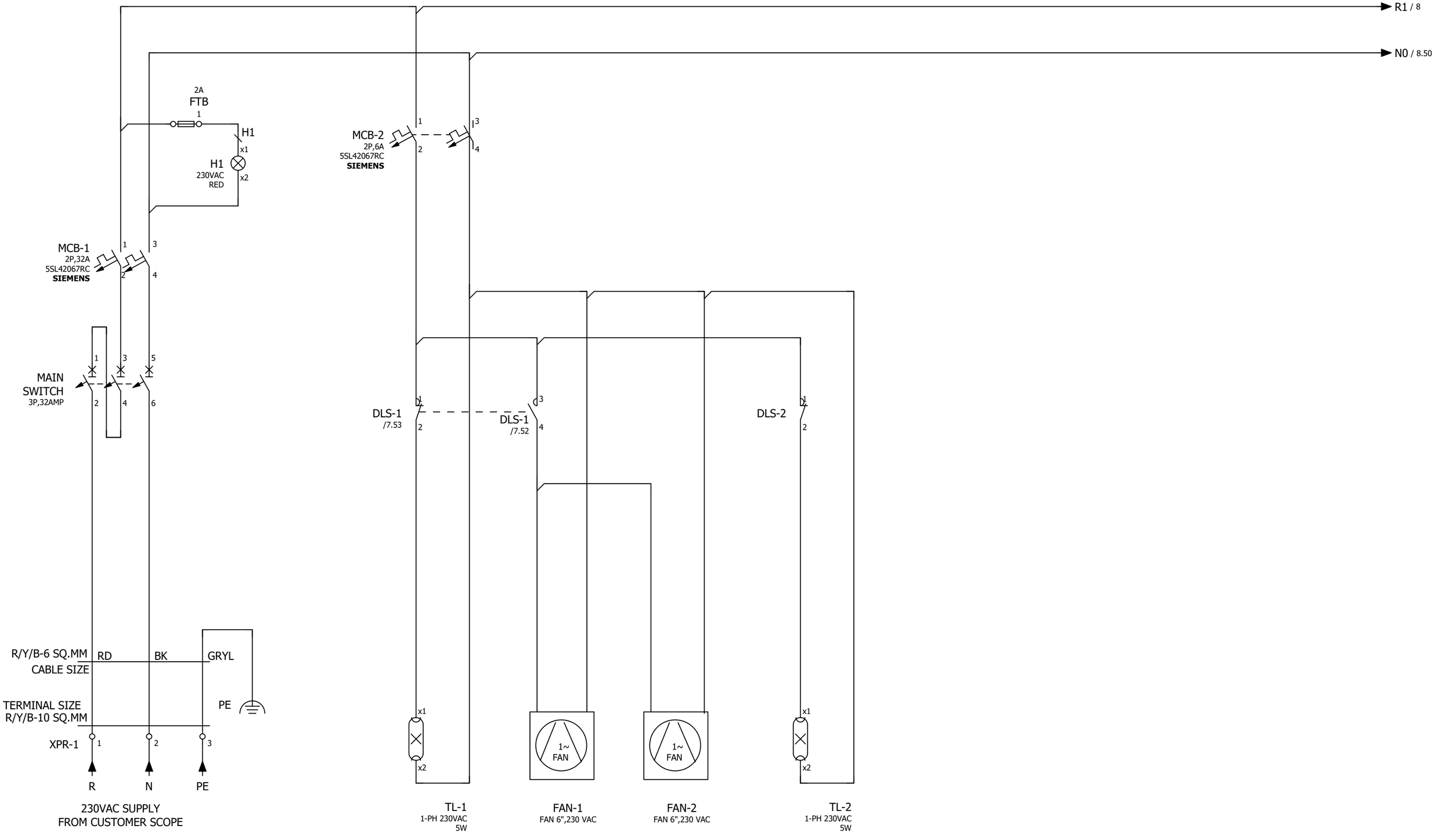
COMPONENT DETAILS:-

CPU	:	CENTRAL PROCESSING UNIT
DI1-7	:	16 CH DIGITAL INPUT MODULES
DO1-4	:	16 CH DIGITAL OUTPUT MODULES
AI1-3	:	8 CH ANALOG INPUT MODULES
AO1-2	:	4 CH ANALOG OUTPUT MODULES
ETH	:	ETHERNET SWITCH
DB1-2	:	DIODE MODULE
SMPS1-4	:	230VAC TO 24VDC POWER SUPPLY
RB1-8	:	24VDC 1C/O RELAY BOARD
MCB1-6	:	MCB FOR 230VAC
MCB 21-40	:	MCB FOR 24VDC
P1 & N1	:	230VAC SUPPLY DISTRIBUTION BLOCK
L1/M1 & L2/M2	:	24 VDC SUPPLY DISTRIBUTION BLOCK
1L2,2L2 & 2M2	:	24 VDC FIELD SUPPLY DB
US	:	UTILITY SOCKET
XPR-1	:	INCOMING SUPPLY TERMINAL BLOCK
XTB-1	:	INSTRUMENTS SUPPLY TERMINALS
XDI1-7	:	DIGITAL INPUT TERMINALS
XDO1-4	:	DIGITAL OUTPUT TERMINALS
XAI1-3	:	ANALOG INPUT TERMINALS
XAO1-3	:	ANALOG OUTPUT TERMINALS
PE	:	POWER EARTHING BUSBAR
SMPS 1 & 2	:	10 AMP 24VDC
SMPS 3 & 4	:	20 AMP 24VDC

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230VAC POWER SUPPLY 50Hz



MOJJ ENGINEERING SYSTEMS LTD	03	AS BUILT	28-02-2025	TITLE:- INCOMER WIRING	PROJECT:- SPRAY DRYER
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230VAC POWER SUPPLY 50Hz

7 / R1

7.59 / N0

RD/2.5 BK/2.5

MCB-3
2P,6A
5SL42067RC
SIEMENS

MCB-4
2P,6A
5SL42067RC
SIEMENS

MCB-5
2P,6A
5SL42067RC
SIEMENS

MCB-6
2P,6A
5SL42067RC
SIEMENS

SMPS-1
10A,24VDC
POWER SUPPLY
SIEMENS

SMPS-2
10A,24VDC
POWER SUPPLY
SIEMENS

SMPS-3
20A,24VDC
POWER SUPPLY
SIEMENS

SMPS-4
20A,24VDC
POWER SUPPLY
SIEMENS

SMPS L N V+ V- PE

2A
2 FTB
3 FTB
4 FTB
5 FTB

H2 230VAC GREEN
H3 230VAC GREEN
H4 230VAC GREEN
H5 230VAC GREEN
x1 x2

MCB-21
2P,10A
5SL42107RC
SIEMENS

MCB-22
2P,10A
5SL42107RC
SIEMENS

MCB-23
2P,20A
5SL42207RC
SIEMENS

MCB-24
2P,20A
5SL42207RC
SIEMENS

DB1
20A,24VDC
WEIDMULLER
DIODE ONLY MODULE-1

DB2
40A,24VDC
WEIDMULLER
DIODE ONLY MODULE-2

MCB-25
2P,20A
5SL42207RC
SIEMENS
OG/4 WH/4
L1 M1

MCB-26
2P,40A
5SL42407RC
SIEMENS
OG/6 WH/6
L2 M2

L1 8
M1 8.57

TO CPU HMI &
ETHERNET SWITCH
CONTROL SUPPLY

L2 9
M2 9.5

TO DI/DO
AI/AO

24 VDC POWER SUPPLY

8.51 / L1 → L1 / 9

8.52 / M1 → M1 / 9.0

1mm² OG
1mm² WH

MCB-27
1P,6A
5SL41067RC
SIEMENS

MCB-28
1P,6A
5SL41067RC
SIEMENS

HMI
24V +DC
24V -DC
2L1 2M1
INPUT 24VDC
IE
GND
PANEL VIEW-800

SUPPLY TO HMI

ETH
3L1 M 3M1

24VDC SUPPLY
ETHERNET SWITCH

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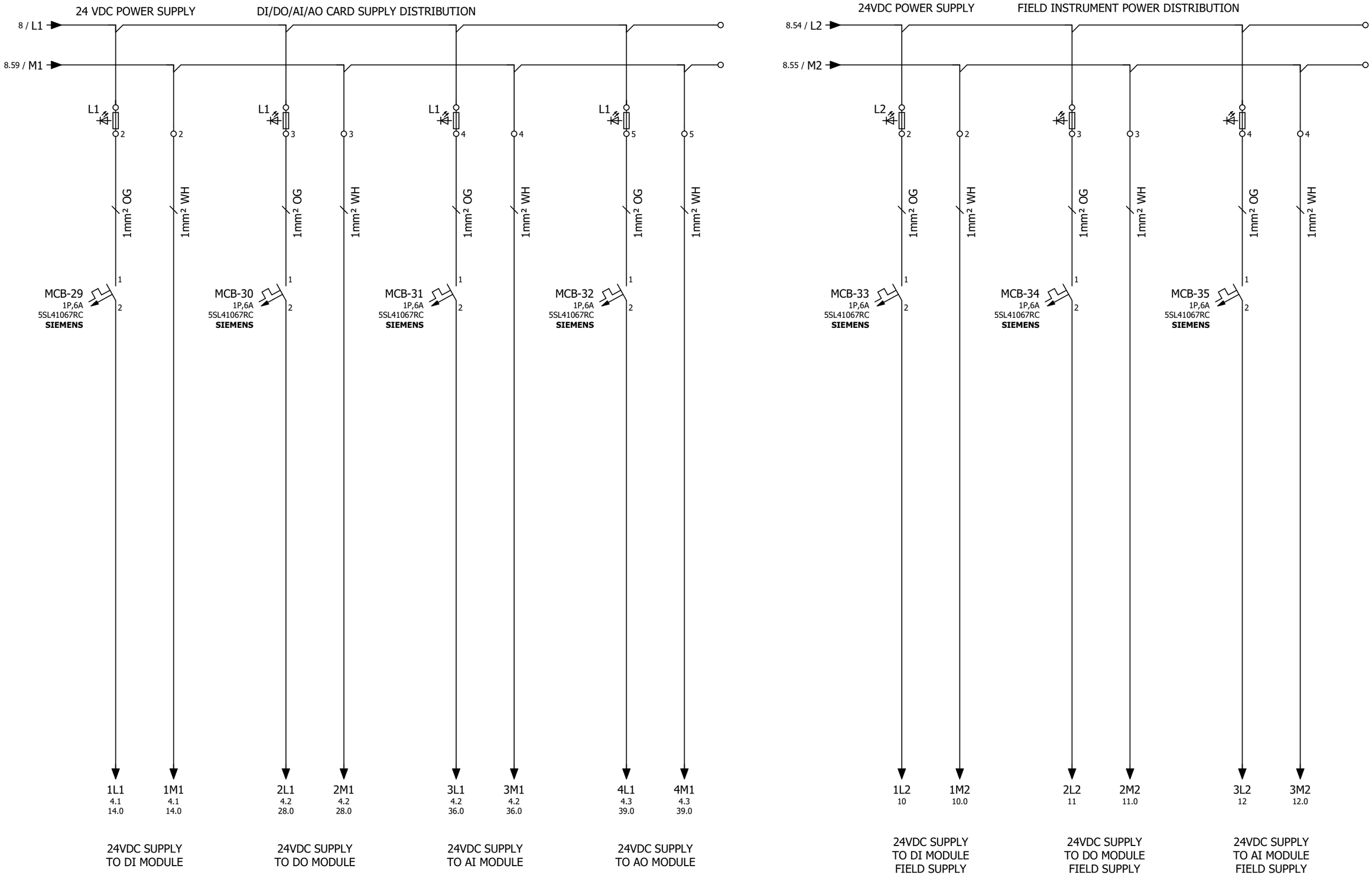
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TITLE:- 24VDC DISTRIBUTION

FILE NO: MES / E&I / MSD-697

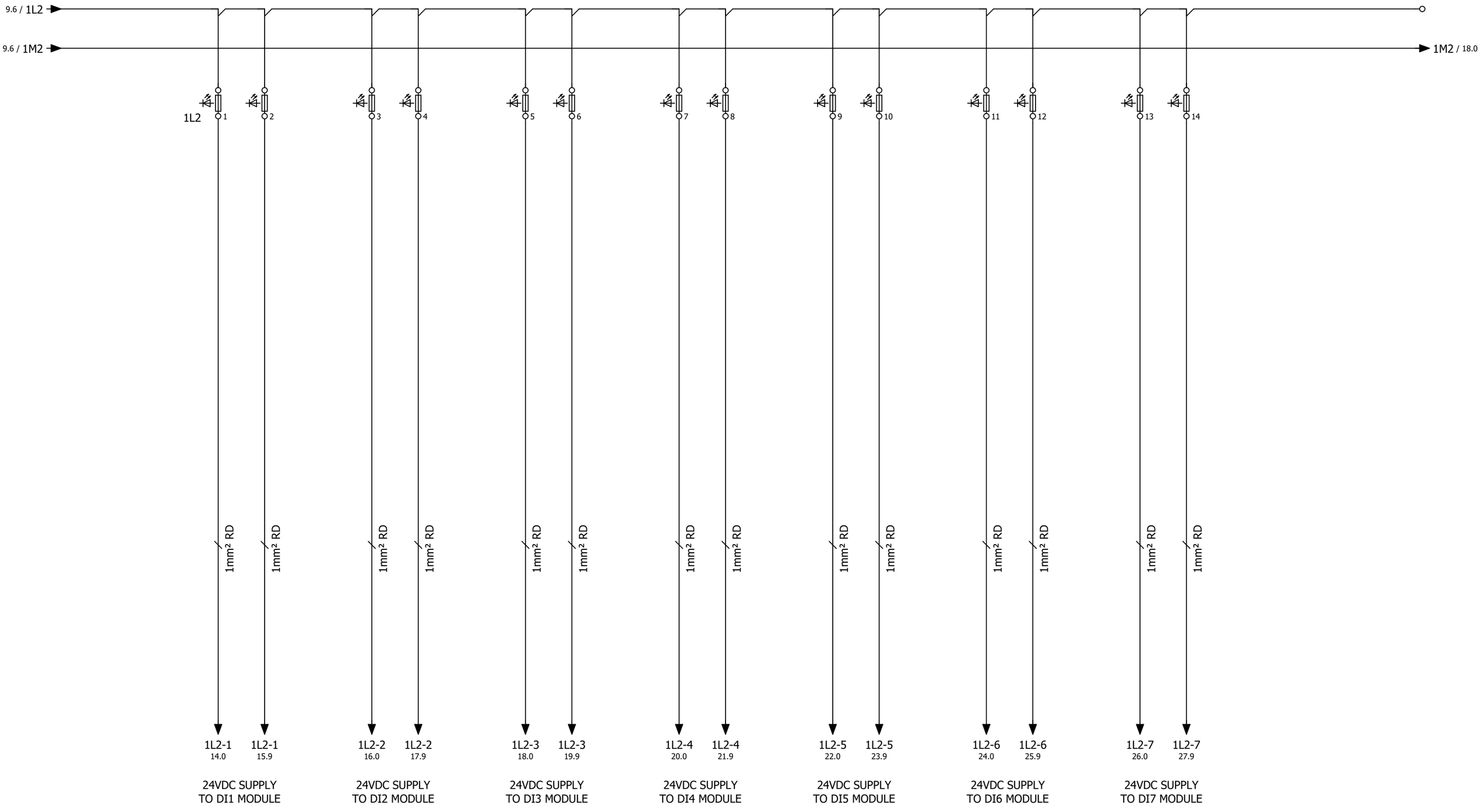
PROJECT:- SPRAY DRYER

PROJECT:- MSD-697

DRG. NO.:-	SHEET NO.	REV.NO.
2024/PRG/MSD/697	10 / 43	AS BUILT

24VDC POWER SUPPLY

DIGITAL INPUT FIELD POWER DISTRIBUTION



MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 24VDC DISTRIBUTION OF DIGITAL INSTRUMENT

FILE NO: MES / E&I / MSD-697

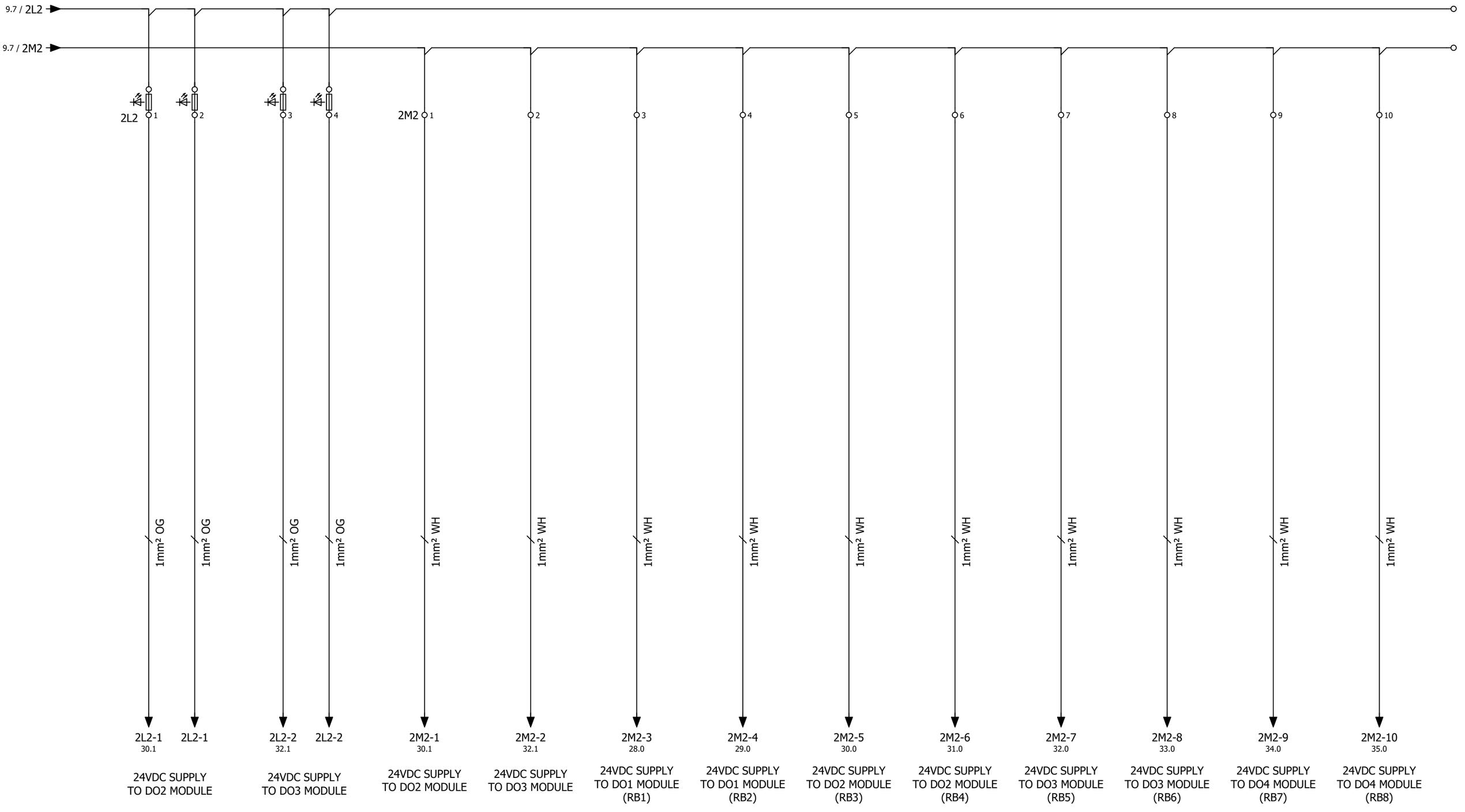
PROJECT:- SPRAY DRYER

PROJECT:- MSD-697

DRG. NO.:-	SHEET NO.	REV.NO.
2024/PRG/MSD/697	11 / 43	AS BUILT

24VDC POWER SUPPLY

DIGITAL OUTPUT FIELD POWER DISTRIBUTION



MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 24VDC DISTRIBUTION OF DIGITAL OUTPUTS

FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER

PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697

SHEET NO. 12 / 43

REV.NO. AS BUILT

24VDC POWER SUPPLY

ANALOG INSTRUMENT FIELD POWER DISTRIBUTION

9.8 / 3L2

9.8 / 3M2

MCB-36
1P,6A
5SL41067RC
SIEMENS

MCB-37
1P,6A
5SL41067RC
SIEMENS

MCB-38
1P,6A
5SL41067RC
SIEMENS

MCB-39
1P,6A
5SL41067RC
SIEMENS

MCB-40
1P,6A
5SL41067RC
SIEMENS

1mm² WH

1mm² OG

1mm² WH

1mm² OG

1mm² WH

1mm² OG

1mm² WH

1mm² OG

1mm² WH

1mm² OG

1mm² WH

XTB-2

1 2 3 4 5 6 7 8 9 10

3M2
38.0

24VDC SUPPLY
TO AI2 MODULE

24VDC SUPPLY
FOR ISNTRUMENTS

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 24VDC DISTRIBUTION OF ANALOG INSTRUMENTS

FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER

PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697

SHEET NO. 13 / 43

REV.NO. AS BUILT

PLC WIRING DIAGRAM

MOJJ ENGINEERING SYSTEMS LTD	03	AS BUILT	28-02-2025	TITLE:- PLC WIRING DIAGRAM			PROJECT:- SPRAY DRYER
	02	AS PER COMMENTS	21-01-2025	FILE NO:	MES / E&I / MSD-697		
	01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.		PROJECT:- MSD-697
	00	FOR APPROVAL	25-11-2024	CHECKED	P.B.		
	REVISION	DETAILS	DATE	APPROVED	MESL		DRG. NO.:- 2024/PRG/MSD/697 / SHEET NO. 14 / 43 / REV.NO. AS BUILT

DI1 | 16 DI DIGITAL INPUT MODULE-1 | 6ES71316BH010BA0 | DI 16x24VDC | 1 OF 2 | SIEMENS

CH0 CH1 CH2 CH3 CH4 CH5 CH6 CH7

L+ M I0.0 I0.1 I0.2 I0.3 I0.4 I0.5 I0.6 I0.7

1 2 3 4 5 6 7 8

9.1 / 1L1

9.1 / 1M1

10.1 / 1L2-1

1L2-1 / 15.0

OG/0.5 BL/0.5 OG/0.5 BL/0.5 BL/0.5 BL/0.5 OG/0.5 BL/0.5 BL/0.5 OG/0.5 BL/0.5

I0:0/0 I0:0/1 I0:0/2 I0:0/3 I0:0/4 I0:0/5 I0:0/6 I0:0/7

500mA 500mA 500mA 500mA 500mA 500mA 500mA 500mA

X-DI1 1 2 3 4 5 6 7 8 9 10 11 12

PANEL SECTION

FIELD SECTION

TAG :	PU-02-O/F	PU-02-T/F	BL-04-A/F	BL-04-O/F	BL-04-T/F	BL-01-O/F	BL-01-T/F	BL-02-O/F
DESC :	SCRUBBER PUMP ON F/B	SCRUBBER PUMP TRIP F/B	ID BLOWER AUTO F/B	ID BLOWER ON F/B	ID BLOWER TRIP F/B	FD BLOWER ON F/B	FD BLOWER TRIP F/B	IBFD BLOWER ON F/B

NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL 500mA

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DI-DIGITAL INPUT WIRING-1

FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER

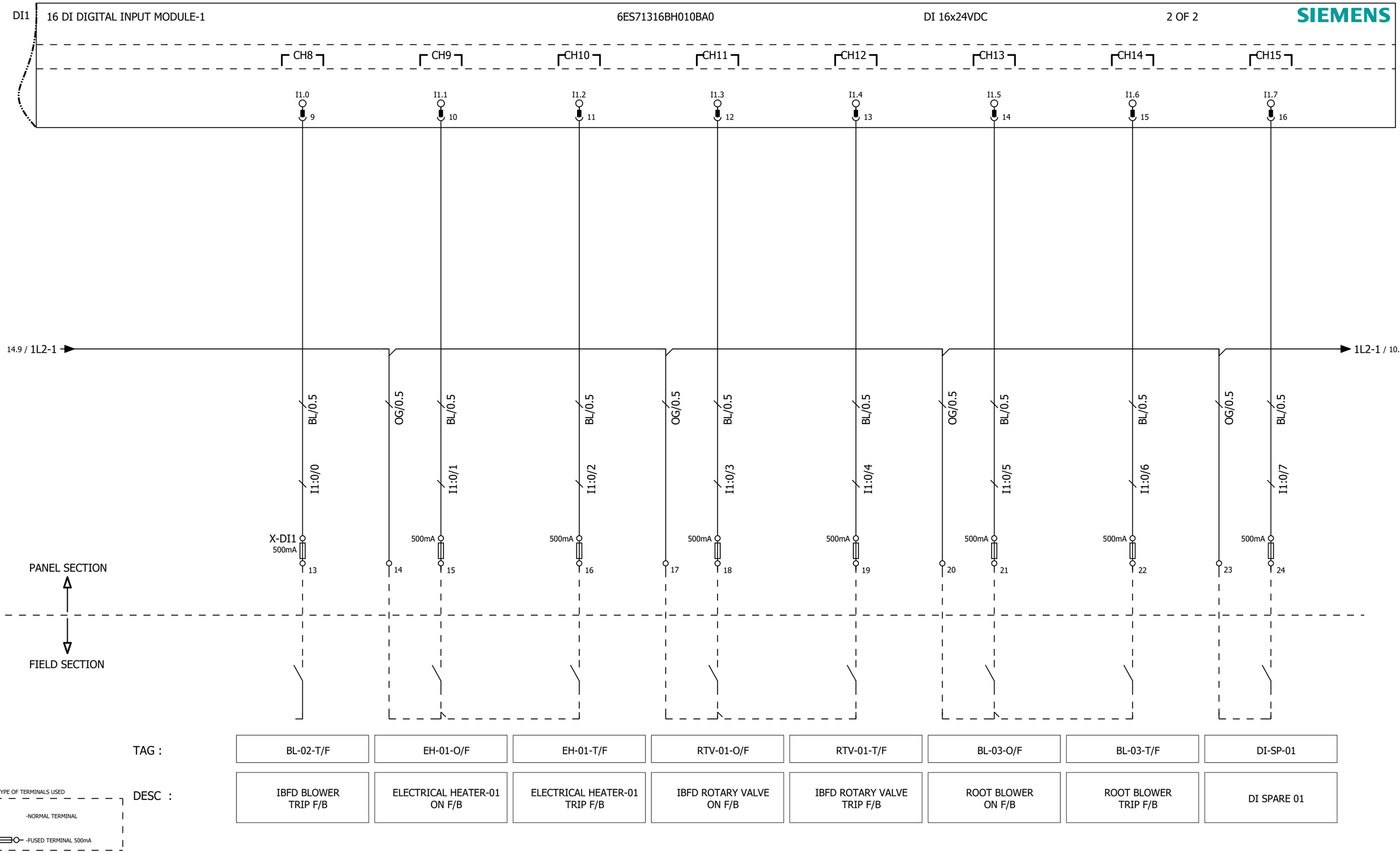
PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697

SHEET NO. 15 / 43

REV.NO. AS BUILT

DI1 | 16 DI DIGITAL INPUT MODULE-1 | 6ES71316BH010BA0 | DI 16x24VDC | 2 OF 2 | SIEMENS



TAG :	BL-02-T/F	EH-01-O/F	EH-01-T/F	RTV-01-O/F	RTV-01-T/F	BL-03-O/F	BL-03-T/F	DI-SP-01
DESC :	IBFD BLOWER TRIP F/B	ELECTRICAL HEATER-01 ON F/B	ELECTRICAL HEATER-01 TRIP F/B	IBFD ROTARY VALVE ON F/B	IBFD ROTARY VALVE TRIP F/B	ROOT BLOWER ON F/B	ROOT BLOWER TRIP F/B	DI SPARE 01

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 500mA

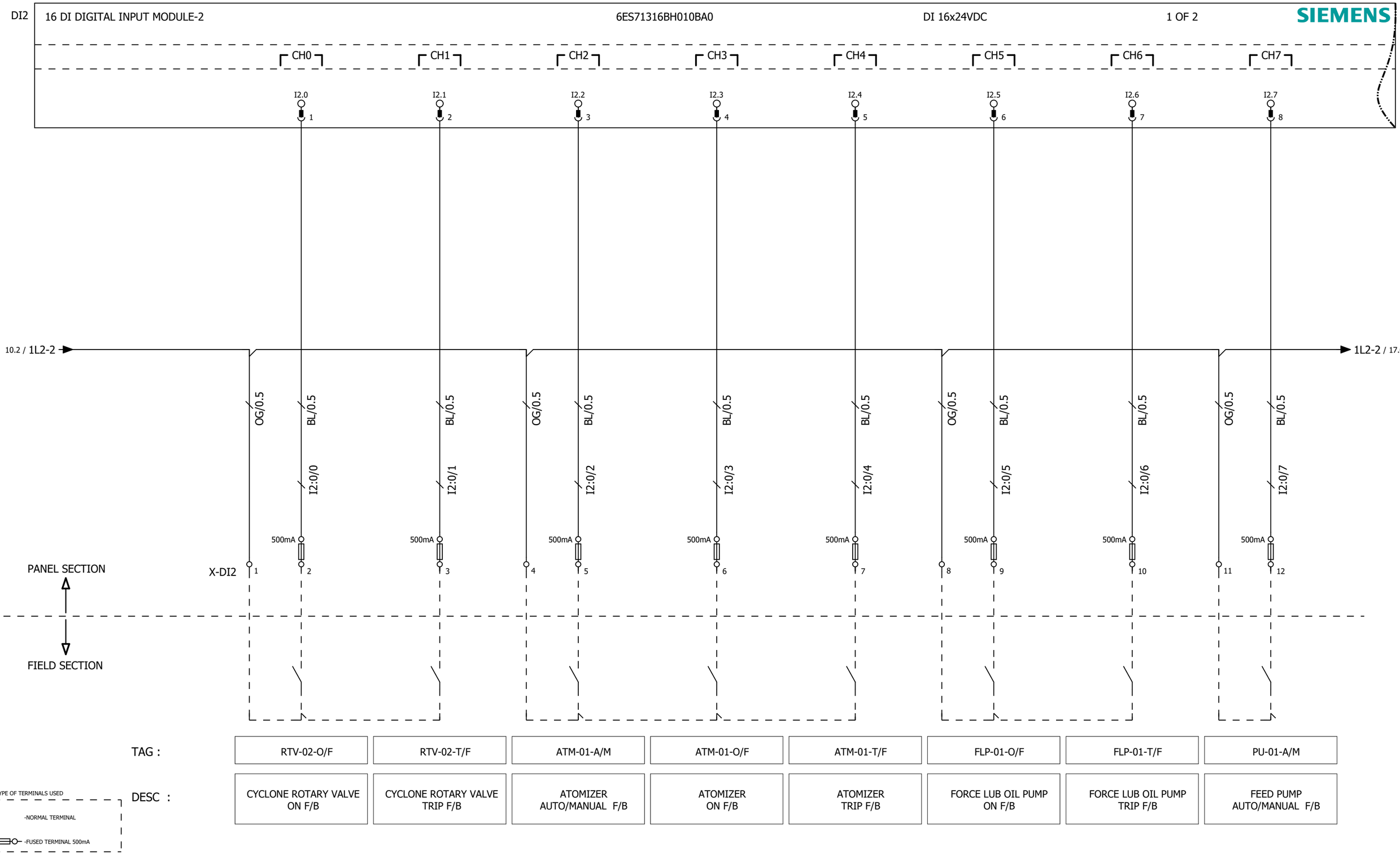
MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DI-DIGITAL INPUT WIRING-1
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697
DRG. NO.:- 2024/PRG/MSD/697
SHEET NO. 16 / 43
REV.NO. AS BUILT

DI2 | 16 DI DIGITAL INPUT MODULE-2 | 6ES71316BH010BA0 | DI 16x24VDC | 1 OF 2 | SIEMENS



TAG :	RTV-02-O/F	RTV-02-T/F	ATM-01-A/M	ATM-01-O/F	ATM-01-T/F	FLP-01-O/F	FLP-01-T/F	PU-01-A/M
DESC :	CYCLONE ROTARY VALVE ON F/B	CYCLONE ROTARY VALVE TRIP F/B	ATOMIZER AUTO/MANUAL F/B	ATOMIZER ON F/B	ATOMIZER TRIP F/B	FORCE LUB OIL PUMP ON F/B	FORCE LUB OIL PUMP TRIP F/B	FEED PUMP AUTO/MANUAL F/B

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 500mA

MOJJ ENGINEERING SYSTEMS LTD

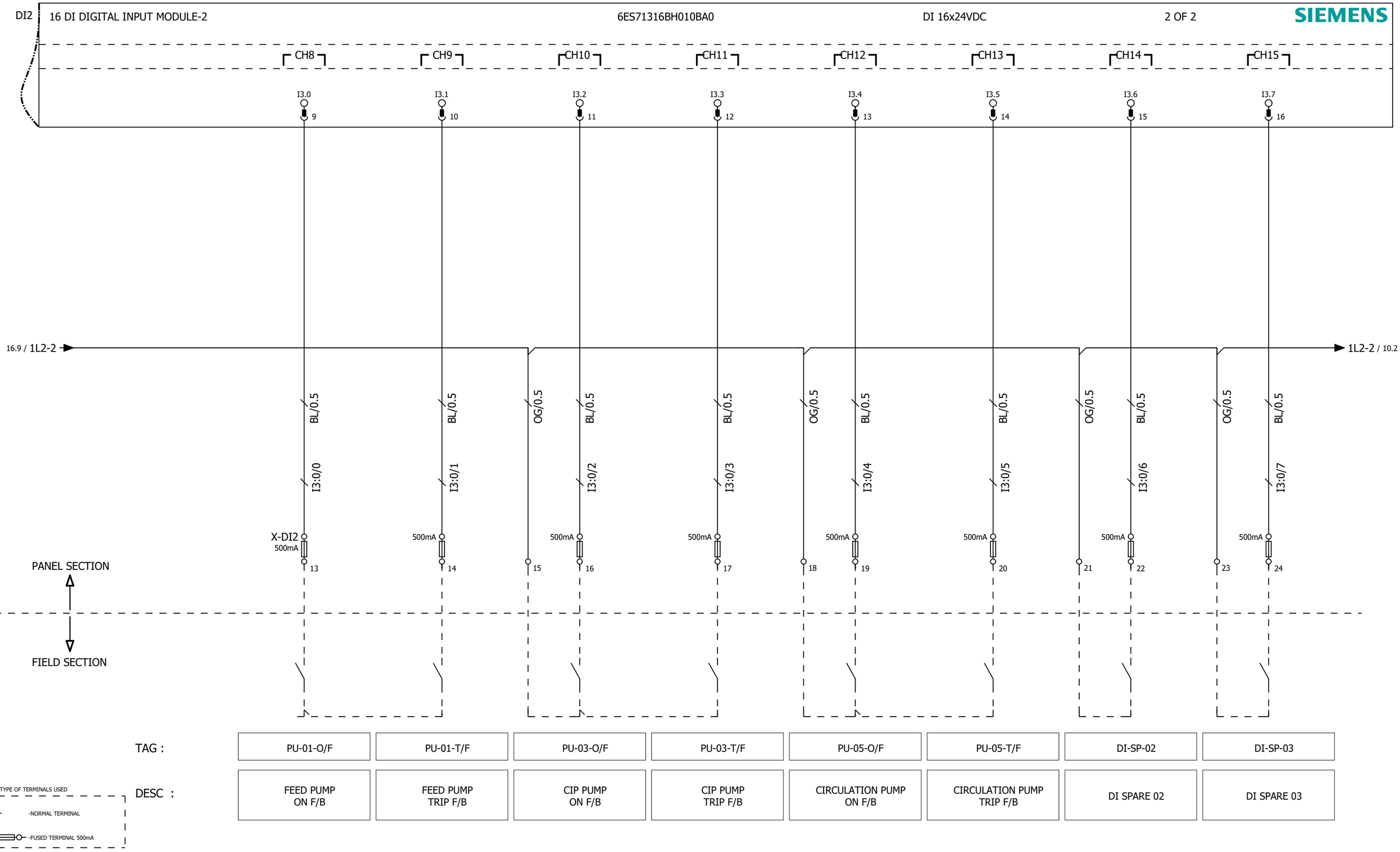
REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DI-DIGITAL INPUT WIRING-2
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697

DRG. NO.:-	SHEET NO.	REV.NO.
2024/PRG/MSD/697	17 / 43	AS BUILT

DI2 | 16 DI DIGITAL INPUT MODULE-2 | 6ES71316BH010BA0 | DI 16x24VDC | 2 OF 2 | SIEMENS



TAG :	DESC :
PU-01-O/F	FEED PUMP ON F/B
PU-01-T/F	FEED PUMP TRIP F/B
PU-03-O/F	CIP PUMP ON F/B
PU-03-T/F	CIP PUMP TRIP F/B
PU-05-O/F	CIRCULATION PUMP ON F/B
PU-05-T/F	CIRCULATION PUMP TRIP F/B
DI-SP-02	DI SPARE 02
DI-SP-03	DI SPARE 03

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 500mA

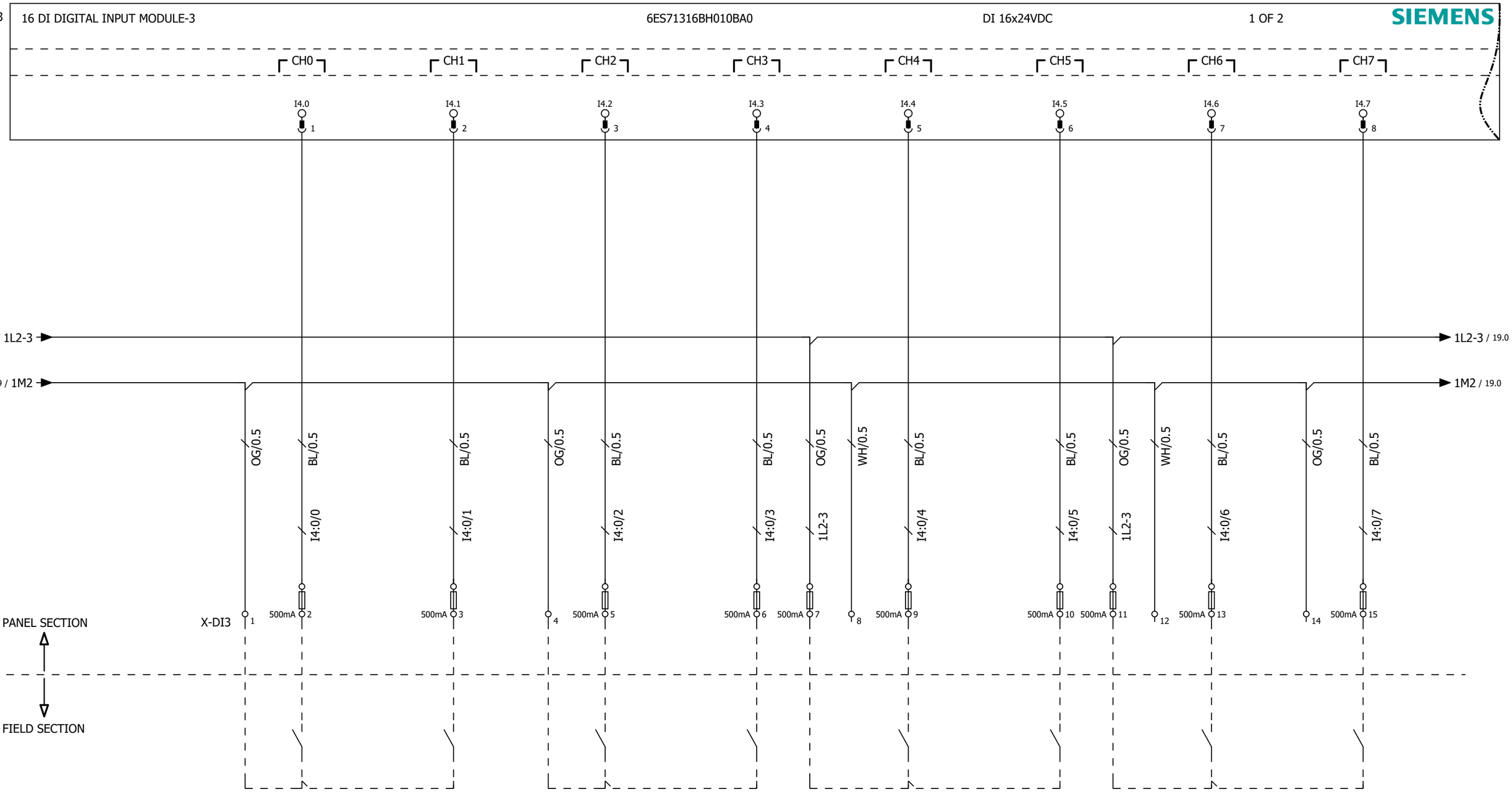
MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DI-DIGITAL INPUT WIRING-2
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697
DRG. NO.:- 2024/PRG/MSD/697
SHEET NO. 18 / 43
REV.NO. AS BUILT

DI3 | 16 DI DIGITAL INPUT MODULE-3 | 6ES71316BH010BA0 | DI 16x24VDC | 1 OF 2 | SIEMENS



TAG :	DESC :
SP-01-O/F	SPARE-01 ON F/B
SP-01-T/F	SPARE-01 TRIP F/B
SP-02-O/F	SPARE-02 ON F/B
SP-02-T/F	SPARE-02 TRIP F/B
ABV-01 O/F	WATER TANK (TK-02) WATER INLET VALVE OPEN F/B
ABV-01 C/F	WATER TANK (TK-02) WATER INLET VALVE CLOSE F/B
ABV-02 O/F	CHAMBER CIP INLET VALVE OPEN F/B
ABV-02 C/F	CHAMBER CIP INLET VALVE CLOSE F/B

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 500mA

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DI-DIGITAL INPUT WIRING-3
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697
DRG. NO.:- 2024/PRG/MSD/697
SHEET NO. 19 / 43
REV.NO. AS BUILT

0	1	2	3	4	5	6	7	8	9

DI3 | 16 DI DIGITAL INPUT MODULE-3 | 6ES71316BH010BA0 | DI 16x24VDC | 2 OF 2 | SIEMENS

CH8 CH9 CH10 CH11 CH12 CH13 CH14 CH15

I5.0 9 | I5.1 10 | I5.2 11 | I5.3 12 | I5.4 13 | I5.5 14 | I5.6 15 | I5.7 16

18.9 / 1L2-3 → 1L2-3 / 10.3

18.9 / 1M2 → 1M2 / 20.0

OG/0.5 1L2-3	WH/0.5	BL/0.5 I5:0/0	BL/0.5 I5:0/1	OG/0.5 1L2-3	WH/0.5	BL/0.5 I5:0/2	BL/0.5 I5:0/3
OG/0.5 1L2-3	WH/0.5	BL/0.5 I5:0/4	BL/0.5 I5:0/5	OG/0.5 1L2-3	WH/0.5	BL/0.5 I5:0/6	BL/0.5 I5:0/7

X-DI3 500mA

16 | 17 | 500mA 18 | 500mA 19 | 500mA 20 | 21 | 500mA 22 | 500mA 23 | 500mA 24 | 25 | 500mA 26 | 500mA 27 | 500mA 28 | 29 | 500mA 30 | 500mA 31

PANEL SECTION

FIELD SECTION

TAG :	DESC :
ABV-03 O/F	CONVEYING LINE DAMPER VALVE OPEN F/B
ABV-03 C/F	CONVEYING LINE DAMPER VALVE CLOSE F/B
ABV-04 O/F	POWER DUCT (CYCLONE INLET) CIP INLET VALVE OPEN F/B
ABV-04 C/F	POWER DUCT (CYCLONE INLET) CIP INLET VALVE CLOSE F/B
ABV-05 O/F	CYCLONE CIP INLET VALVE OPEN F/B
ABV-05 C/F	CYCLONE CIP INLET VALVE CLOSE F/B
ABV-06 O/F	CIP TANK OUTLET VALVE OPEN F/B
ABV-06 C/F	CIP TANK OUTLET VALVE CLOSE F/B

NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL 500mA

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
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00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DI-DIGITAL INPUT WIRING-3

FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER

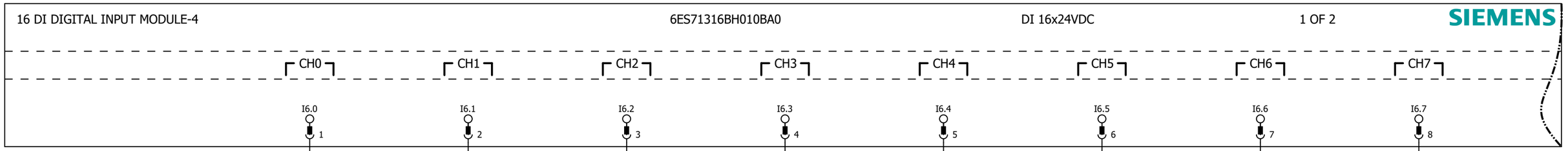
PROJECT:- MSD-697

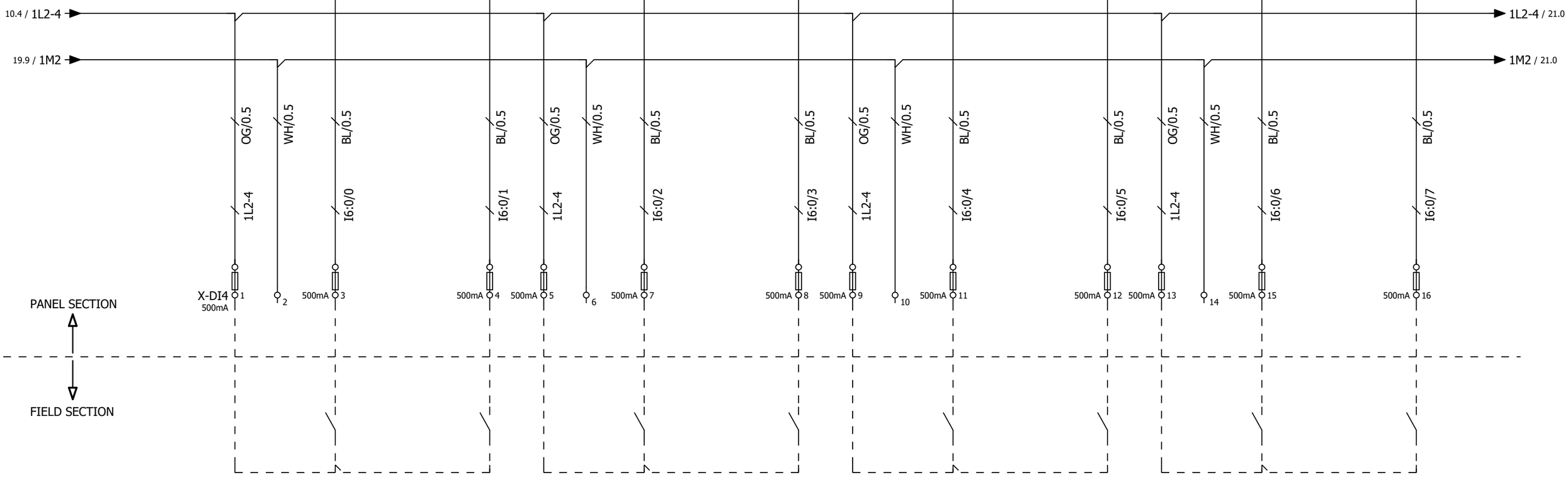
DRG. NO.:- 2024/PRG/MSD/697

SHEET NO. 20 / 43

REV.NO. AS BUILT

DI4 | 16 DI DIGITAL INPUT MODULE-4 | 6ES71316BH010BA0 | DI 16x24VDC | 1 OF 2 | SIEMENS





TAG :	ABV-07 O/F	ABV-07 C/F	ABV-08 O/F	ABV-08 C/F	ABV-09 O/F	ABV-09 C/F	ABV-10 O/F	ABV-10 C/F
DESC :	CIP LINE DRAIN VALVE OPEN F/B	CIP LINE DRAIN VALVE CLOSE F/B	WINCH CIP INLET VALVE OPEN F/B	WINCH CIP INLET VALVE CLOSE F/B	SCRUBBER TANK DRAIN VALVE OPEN F/B	SCRUBBER TANK DRAIN VALVE CLOSE F/B	SCRUBBER TANK WATER INLET VALVE OPEN F/B	SCRUBBER TANK WATER INLET VALVE CLOSE F/B

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 500mA

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DI-DIGITAL INPUT WIRING-4
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697
DRG. NO.:- 2024/PRG/MSD/697
SHEET NO. 21 / 43
REV.NO. AS BUILT

0	1	2	3	4	5	6	7	8	9

DI4 | 16 DI DIGITAL INPUT MODULE-4 | 6ES71316BH010BA0 | DI 16x24VDC | 2 OF 2 | SIEMENS

CH8 CH9 CH10 CH11 CH12 CH13 CH14 CH15

I7.0 9 | I7.1 10 | I7.2 11 | I7.3 12 | I7.4 13 | I7.5 14 | I7.6 15 | I7.7 16

20.9 / 1L2-4 → 1L2-4 / 10.4

20.9 / 1M2 → 1M2 /

OG/0.5 1L2-4	WH/0.5	BL/0.5 I7:0/0	BL/0.5 I7:0/1	OG/0.5 1L2-4	WH/0.5	BL/0.5 I7:0/2	BL/0.5 I7:0/3
X-DI4 17 500mA	18	500mA 19	500mA 20	500mA 21	22	500mA 23	500mA 24

WH/0.5	BL/0.5 I7:0/4	BL/0.5 I7:0/5	OG/0.5	BL/0.5 I7:0/6	BL/0.5 I7:0/7		
25	500mA 26	500mA 27	28	500mA 29	500mA 30		

PANEL SECTION

FIELD SECTION

TAG :	ABV-11 O/F	ABV-11 C/F	ABV-12 O/F	ABV-12 C/F	ABV-13 O/F	ABV-13 C/F	ABV-14 O/F	ABV-14 C/F
DESC :	CYCLONE CIP OUTLET VALVE OPEN F/B	CYCLONE CIP OUTLET VALVE CLOSE F/B	CHAMBER CIP OUTLET VALVE OPEN F/B	CHAMBER CIP OUTLET VALVE CLOSE F/B	CIP TANK RETURN CIP INLET VALVE OPEN F/B	CIP TANK RETURN CIP INLET VALVE CLOSE F/B	FEED TANK CIP INLET VALVE OPEN F/B	FEED TANK CIP INLET VALVE CLOSE F/B

NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL 500mA

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
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01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DI-DIGITAL INPUT WIRING-4

FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER

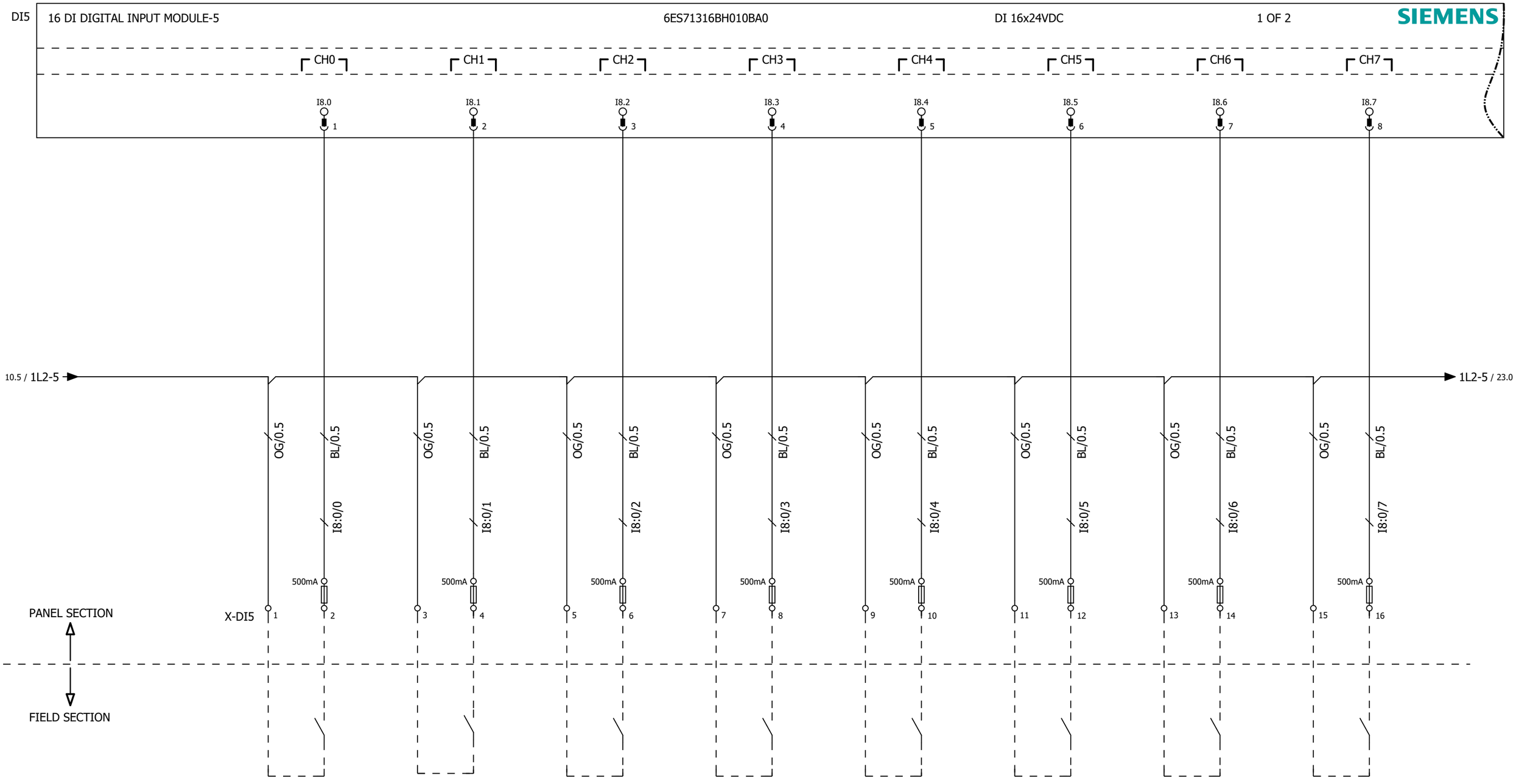
PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697

SHEET NO. 22 / 43

REV.NO. AS BUILT

DI5 | 16 DI DIGITAL INPUT MODULE-5 | 6ES71316BH010BA0 | DI 16x24VDC | 1 OF 2 | SIEMENS



TAG :	ABV-15 O/F	ABV-15 C/F	ABV-16 O/F	ABV-16 C/F	ADV-01 O/F	ADV-01 C/F	ADV-02 O/F	ADV-02 C/F
DESC :	CIP VALVE-1 OPEN F/B	CIP VALVE-1 CLOSE F/B	CIP TANK FRESH WATER INLET VALVE OPEN F/B	CIP TANK FRESH WATER INLET VALVE CLOSE F/B	FEED TANK FEED INLET VALVE OPEN F/B	FEED TANK FEED INLET VALVE CLOSE F/B	FEED TANK FEED OUTLET VALVE OPEN F/B	FEED TANK FEED OUTLET VALVE CLOSE F/B

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 500mA

MOJJ ENGINEERING SYSTEMS LTD

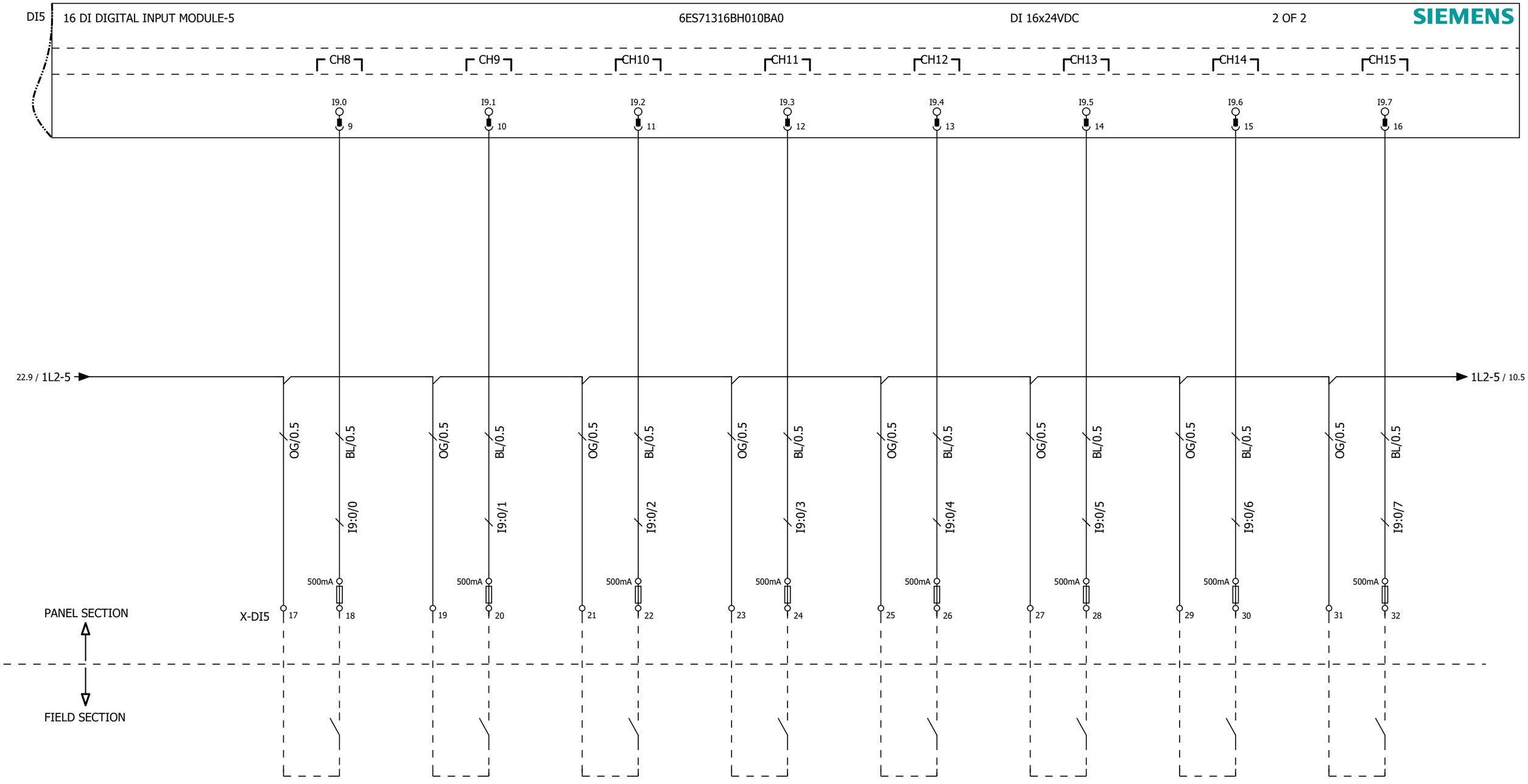
REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DI-DIGITAL INPUT WIRING-5
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697

DRG. NO.:-	SHEET NO.	REV.NO.
2024/PRG/MSD/697	23 / 43	AS BUILT

DI5 | 16 DI DIGITAL INPUT MODULE-5 | 6ES71316BH010BA0 | DI 16x24VDC | 2 OF 2 | SIEMENS



TAG :	ADV-03 O/F	ADV-03 C/F	AGV-01 O/F	AGV-01 C/F	BR L F/B	BR R F/B	BR IH F/B	BR-RUN-F/B
DESC :	WATER TANK OUTLET VALVE OPEN F/B	WATER TANK OUTLET VALVE CLOSE F/B	CIP TANK STEAM INLET VALVE OPEN F/B	CIP TANK STEAM INLET VALVE CLOSE F/B	BURNER LOCAL SELECTION F/B	BURNER REMOTE SELECTION F/B	BURNER INTERLOCK HEALTHY F/B	BURNER RUN F/B

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 500mA

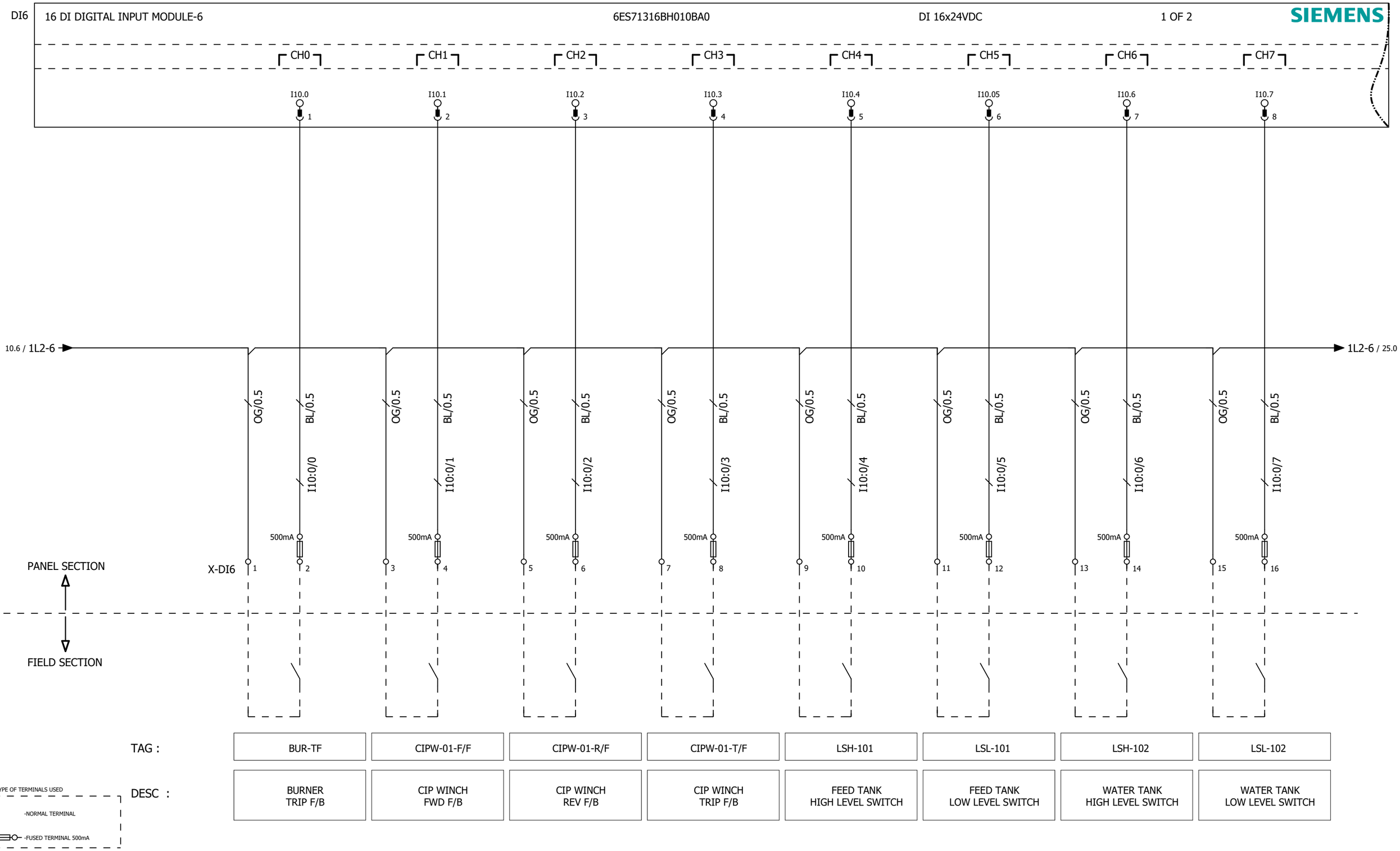
MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DI-DIGITAL INPUT WIRING-5
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697
DRG. NO.:- 2024/PRG/MSD/697
SHEET NO. 24 / 43
REV.NO. AS BUILT

DI6 | 16 DI DIGITAL INPUT MODULE-6 | 6ES71316BH010BA0 | DI 16x24VDC | 1 OF 2 | SIEMENS



TAG :	DESC :
BUR-TF	BURNER TRIP F/B
CIPW-01-F/F	CIP WINCH FWD F/B
CIPW-01-R/F	CIP WINCH REV F/B
CIPW-01-T/F	CIP WINCH TRIP F/B
LSH-101	FEED TANK HIGH LEVEL SWITCH
LSL-101	FEED TANK LOW LEVEL SWITCH
LSH-102	WATER TANK HIGH LEVEL SWITCH
LSL-102	WATER TANK LOW LEVEL SWITCH

NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL 500mA

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DI-DIGITAL INPUT WIRING-6

FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER

PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697

SHEET NO. 25 / 43

REV.NO. AS BUILT

DI6 | 16 DI DIGITAL INPUT MODULE-6 | 6ES71316BH010BA0 | DI 16x24VDC | 2 OF 2 | SIEMENS

CH8 CH9 CH10 CH11 CH12 CH13 CH14 CH15

I11.0 9 | I11.1 10 | I11.2 11 | I11.3 12 | I11.4 13 | I11.5 14 | I11.6 15 | I11.7 16

24.9 / 1L2-6

1L2-6 / 10.6

OG/0.5 BL/0.5 — I11:0/0 — 500mA — X-DI6 17 18

OG/0.5 BL/0.5 — I11:0/1 — 500mA — 19 20

OG/0.5 BL/0.5 — I11:0/2 — 500mA — 21 22

OG/0.5 BL/0.5 — I11:0/3 — 500mA — 23 24

OG/0.5 BL/0.5 — I11:0/4 — 500mA — 25 26

OG/0.5 BL/0.5 — I11:0/5 — 500mA — 27 28

OG/0.5 BL/0.5 — I11:0/6 — 500mA — 29 30

OG/0.5 BL/0.5 — I11:0/7 — 500mA — 31 32

PANEL SECTION

FIELD SECTION

TAG :	DESC :
LSH-103	SRUBBER TANK HIGH LEVEL SWITCH
LSL-103	SRUBBER TANK LOW LEVEL SWITCH
LSH-104	CIP TANK HIGH LEVEL SWITCH
LSL-104	CIP TANK LOW LEVEL SWITCH
PS-101	SCRUBBER PUMP OUTLET PRESSURE SWITCH
LUB_LVL	LUBRICATION OIL LEVEL
MCC-PF	MCC PANEL POWER FAILURE F/B
MCC E-STOP	MCC PANEL EMRGENCY STOP F/B

NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL 500mA

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025	DRAWN BY	S.B.
02	AS PER COMMENTS	21-01-2025	CHECKED	P.B.
01	AS PER COMMENTS	14-01-2025		
00	FOR APPROVAL	25-11-2024		

TITLE:- 16DI-DIGITAL INPUT WIRING-6

FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER

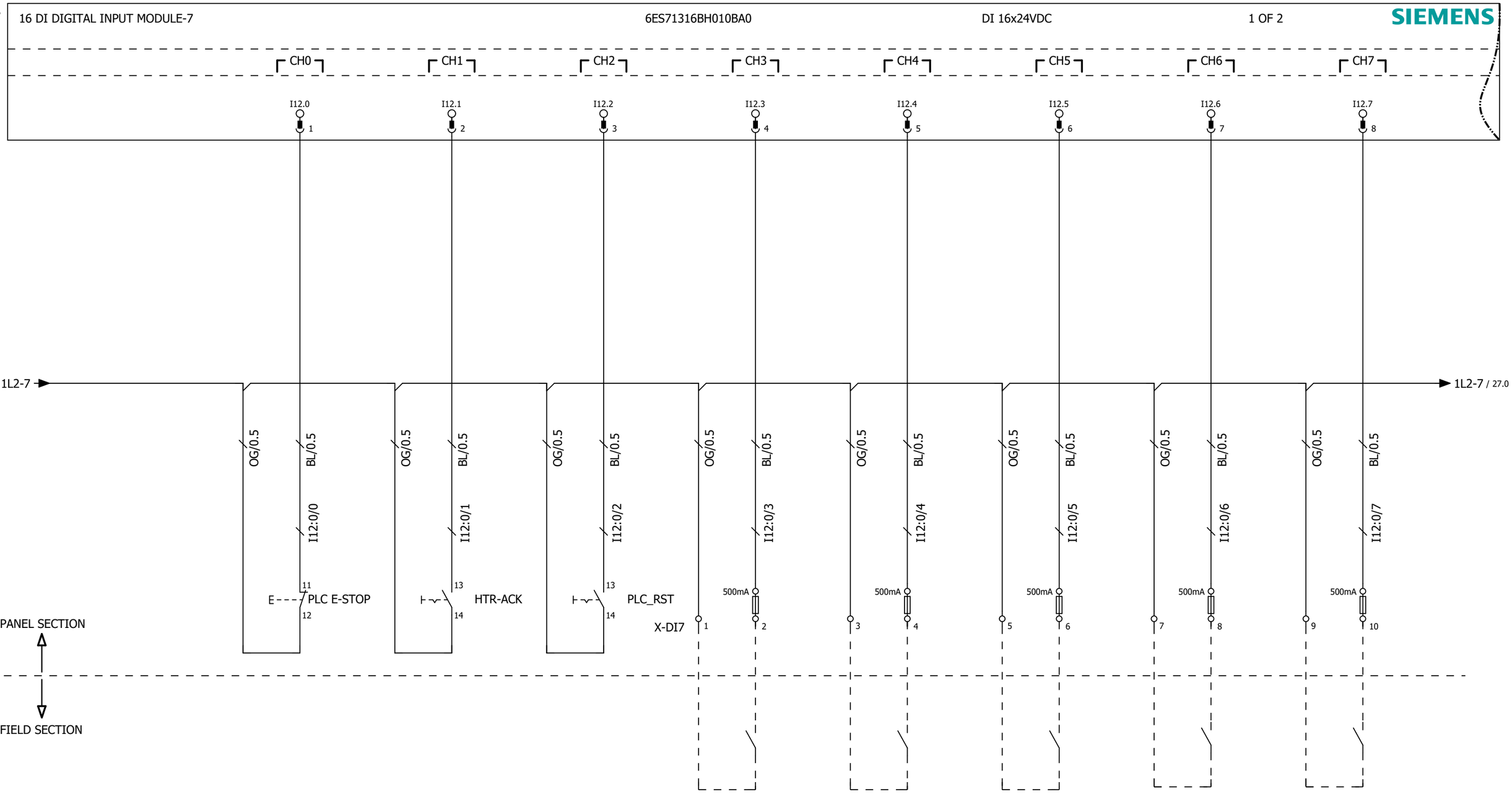
PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697

SHEET NO. 26 / 43

REV.NO. AS BUILT

DI7 | 16 DI DIGITAL INPUT MODULE-7 | 6ES71316BH010BA0 | DI 16x24VDC | 1 OF 2 | SIEMENS



TAG :	PLC E-STOP	HTR-ACK	PLC_RST	RL_FB	DI-SP-04	DI-SP-05	DI-SP-06	DI-SP-07
DESC :	PLC PANEL EMRGENCY STOP F/B	HOOTER ACKNOWLEGEMENT F/B	PLC ALARM RESET F/B	MCC REMOTE/LOCAL F/B	DI SPARE 04	DI SPARE 05	DI SPARE 06	DI SPARE 07

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 500mA

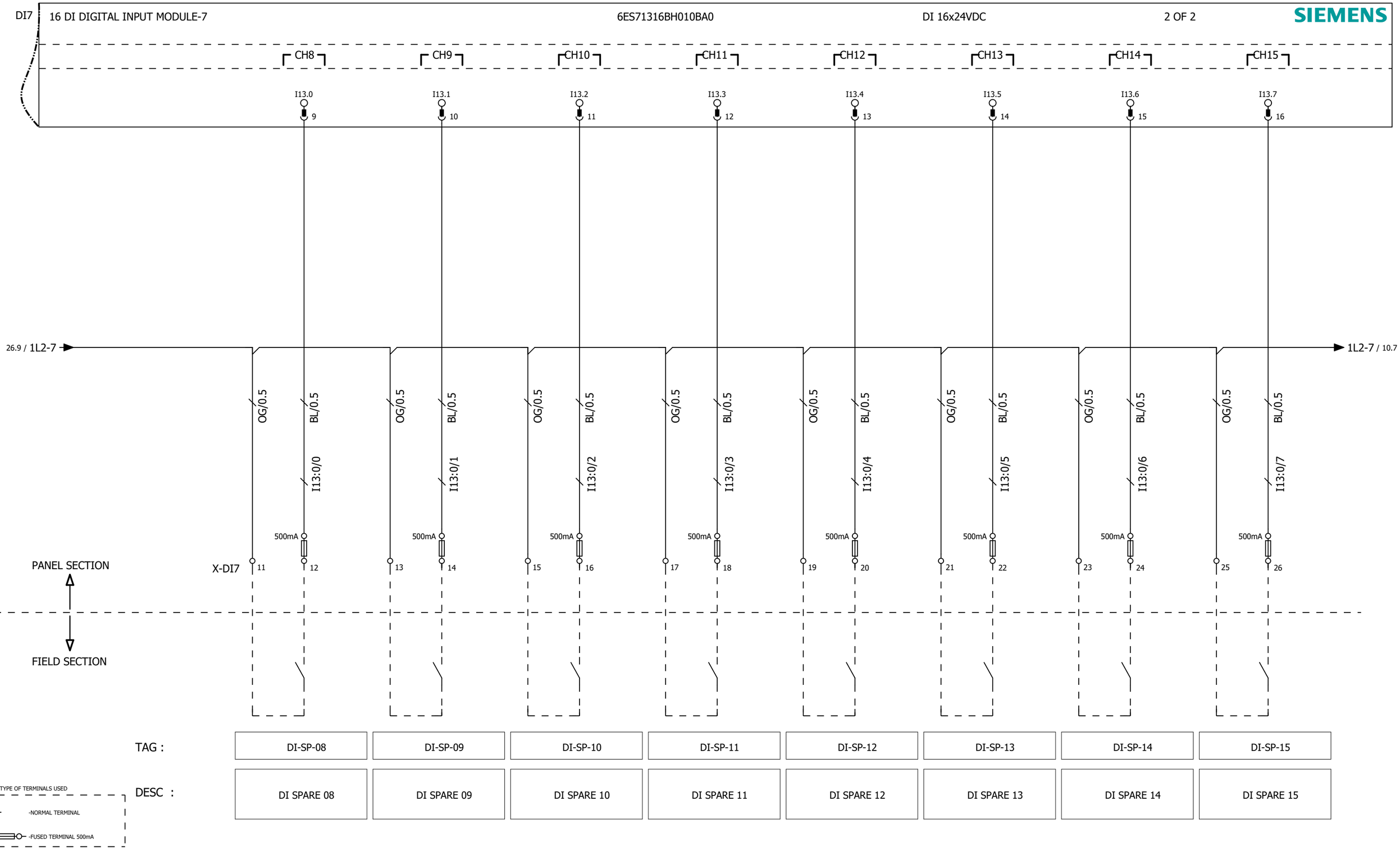
MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DI-DIGITAL INPUT WIRING-7
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697
DRG. NO.:- 2024/PRG/MSD/697
SHEET NO. 27 / 43
REV.NO. AS BUILT

DI7 | 16 DI DIGITAL INPUT MODULE-7 | 6ES71316BH010BA0 | DI 16x24VDC | 2 OF 2 | SIEMENS



TAG :	DI-SP-08	DI-SP-09	DI-SP-10	DI-SP-11	DI-SP-12	DI-SP-13	DI-SP-14	DI-SP-15
DESC :	DI SPARE 08	DI SPARE 09	DI SPARE 10	DI SPARE 11	DI SPARE 12	DI SPARE 13	DI SPARE 14	DI SPARE 15

NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL 500mA

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025	DRAWN BY	S.B.
02	AS PER COMMENTS	21-01-2025	CHECKED	P.B.
01	AS PER COMMENTS	14-01-2025		
00	FOR APPROVAL	25-11-2024		

TITLE:- 16DI-DIGITAL INPUT WIRING-7

FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER

PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697

SHEET NO. 28 / 43

REV.NO. AS BUILT

DO1 | 16 DO DIGITAL OUTPUT MODULE-1 | 6ES71326BH010BA0 | DQ 16x24 VDC | 1 OF 2 | SIEMENS

CH0 | CH1 | CH2 | CH3 | CH4 | CH5 | CH6 | CH7

L+ | M | Q0.0 1 | Q0.1 2 | Q0.2 3 | Q0.3 4 | Q0.4 5 | Q0.5 6 | Q0.6 7 | Q0.7 8

9.2 / 2L1

9.2 / 2M1

11.4 / 2M2-3

Q0:0/0 | Q0:0/1 | Q0:0/2 | Q0:0/3 | Q0:0/4 | Q0:0/5 | Q0:0/6 | Q0:0/7

RELAY BOARD-1
8 CH (RB-01) 24VDC-1C/0

-1 +1	-2 +2	-3 +3	-4 +4	-5 +5	-6 +6	-7 +7	-8 +8
A1	A1	A1	A1	A1	A1	A1	A1
RB01-R1	RB01-R2	RB01-R3	RB01-R4	RB01-R5	RB01-R6	RB01-R7	RB01-R8
A2	A2	A2	A2	A2	A2	A2	A2
P1 C1 O1	P2 C2 O2	P3 C3 O3	P4 C4 O4	P5 C5 O5	P6 C6 O6	P7 C7 O7	P8 C8 O8

PANEL SECTION

FIELD SECTION

X-DO1

2A 1	2	2A 3	4	2A 5	6	2A 7	8	2A 9	10	2A 11	12	2A 13	14	2A 15	16
TAG :	PU-02-CMD		BL-04-CMD		BL-01-CMD		BR-01-CMD		BL-02-CMD		EH-01-CMD		RTV-01-CMD		BL-03-CMD	
DESC :	SCRUBBER PUMP START/STOP COMMAND		ID BLOWER START/STOP COMMAND		FD BLOWER START/STOP COMMAND		BURNER START/STOP COMMAND		IBFD BLOWER START/STOP COMMAND		ELECTRICAL HEATER-01 START/STOP COMMAND		IBFD ROTARY VALVE START/STOP COMMAND		ROOT BLOWER START/STOP COMMAND	

NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL 2A

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025	DRAWN BY	S.B.
02	AS PER COMMENTS	21-01-2025	CHECKED	P.B.
01	AS PER COMMENTS	14-01-2025		
00	FOR APPROVAL	25-11-2024		

TITLE:- 16DO-DIGITAL OUTPUT WIRING-1

FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER

PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697

SHEET NO. 29 / 43

REV.NO. AS BUILT

DO1 16 DO DIGITAL OUTPUT MODULE-1 6ES71326BH010BA0 DQ 16x24 VDC 2 OF 2 SIEMENS

CH8 CH9 CH10 CH11 CH12 CH13 CH14 CH15

Q1.0 Q1.1 Q1.2 Q1.3 Q1.4 Q1.5 Q1.6 Q1.7

1 2 3 4 5 6 7 8

Q1:0/0 Q1:0/1 Q1:0/2 Q1:0/3 Q1:0/4 Q1:0/5 Q1:0/6 Q1:0/7

11.5 / 2M2-4

RELAY BOARD-2
8 CH (RB-02) 24VDC-1C/0

-1 +1 A1 RB02-R1 A2
-2 +2 A1 RB02-R2 A2
-3 +3 A1 RB02-R3 A2
-4 +4 A1 RB02-R4 A2
-5 +5 A1 RB02-R5 A2
-6 +6 A1 RB02-R6 A2
-7 +7 A1 RB02-R7 A2
-8 +8 A1 RB02-R8 A2

P1 C1 O1 P2 C2 O2 P3 C3 O3 P4 C4 O4 P5 C5 O5 P6 C6 O6 P7 C7 O7 P8 C8 O8

PANEL SECTION

FIELD SECTION

X-DO1

2A 17 18 2A 19 20 2A 21 22 2A 23 24 2A 25 26 2A 27 28 2A 29 30 2A 31 32

TAG :	DESC :
RTV-02-CMD	CYCLONE ROTARY VALVE START/STOP COMMAND
ATM-01-CMD	ATOMIZER START/STOP COMMAND
FLP-01-CMD	FORCE LUB OIL PUMP START/STOP COMMAND
PU-01-CMD	FEED PUMP START/STOP COMMAND
PU-03-CMD	CIP PUMP START/STOP COMMAND
PU-05-CMD	CIRCULATION PUMP START/STOP COMMAND
CIPW-01 CMD	CIP WINCH FWD COMMAND
CIPW-01 CMD	CIP WINCH REV COMMAND

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 2A

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DO-DIGITAL OUTPUT WIRING-1
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697
DRG. NO.:- 2024/PRG/MSD/697
SHEET NO. 30 / 43
REV.NO. AS BUILT

DO2 | 16 DO DIGITAL OUTPUT MODULE-2 | 6ES71326BH010BA0 | DQ 16x24 VDC | 1 OF 2 | SIEMENS

CH0 CH1 CH2 CH3 CH4 CH5 CH6 CH7

Q2.0 1 | Q2.1 2 | Q2.2 3 | Q2.3 4 | Q2.4 5 | Q2.5 6 | Q2.6 7 | Q2.7 8

Q2:0/0 | Q2:0/1 | Q2:0/2 | Q2:0/3 | Q2:0/4 | Q2:0/5 | Q2:0/6 | Q2:0/7

11.5 / 2M2-5

RELAY BOARD-3
8 CH (RB-03) 24VDC-1C/0

-1 +1	-2 +2	-3 +3	-4 +4	-5 +5	-6 +6	-7 +7	-8 +8
A1	A1	A1	A1	A1	A1	A1	A1
RB03-R1	RB03-R2	RB03-R3	RB03-R4	RB03-R5	RB03-R6	RB03-R7	RB03-R8
A2	A2	A2	A2	A2	A2	A2	A2
P1 O1 C1	P2 O2 C2	P3 O3 C3	P4 O4 C4	P5 O5 C5	P6 O6 C6	P7 O7 C7	P8 O8 C8

11.1 / 2L2-1 → 2L2-1 / 30

11.2 / 2M2-1 → 2M2-1 / 30

PANEL SECTION

FIELD SECTION

X-DO2 2A: 1 2 | 2A 3 4 | 2A 5 6 | 2A 7 8 | 2A 9 10 | 2A 11 12 | 2A 13 14 | 2A 15 16

TAG :	DESC :
SP-01-CMD	SPARE-01 START/STOP COMMAND
SP-02-CMD	SPARE-02 START/STOP COMMAND
ABV-01 CMD	WATER TANK (TK-02) WATER INLET VALVE OPEN/CLOSE CMD
ABV-02 CMD	CHAMBER CIP INLET VALVE OPEN/CLOSE CMD
ABV-03 CMD	CONVEYING LINE DAMPER VALVE OPEN/CLOSE CMD
ABV-04 CMD	POWDER DUCT(CYCLONE INLET) CIP INLET VALVE OPEN/CLOSE CMD
ABV-05 CMD	CYCLONE CIP INLET VALVE OPEN/CLOSE CMD
ABV-06 CMD	CIP TANK OUTLET VALVE OPEN/CLOSE CMD

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 2A

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DO-DIGITAL OUTPUT WIRING-2

FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER

PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697

SHEET NO. 31 / 43

REV.NO. AS BUILT

DO2 | 16 DO DIGITAL OUTPUT MODULE-2 | 6ES71326BH010BA0 | DQ 16x24 VDC | 2 OF 2 | SIEMENS

CH8 CH9 CH10 CH11 CH12 CH13 CH14 CH15

Q3.0 1 | Q3.1 2 | Q3.2 3 | Q3.3 4 | Q3.4 5 | Q3.5 6 | Q3.6 7 | Q3.7 8

Q3:0/0 | Q3:0/1 | Q3:0/2 | Q3:0/3 | Q3:0/4 | Q3:0/5 | Q3:0/6 | Q3:0/7

11.6 / 2M2-6

RELAY BOARD-4
8 CH (RB-04) 24VDC-1C/0

-1 +1	-2 +2	-3 +3	-4 +4	-5 +5	-6 +6	-7 +7	-8 +8
A1	A1	A1	A1	A1	A1	A1	A1
RB04-R1	RB04-R2	RB04-R3	RB04-R4	RB04-R5	RB04-R6	RB04-R7	RB04-R8
A2	A2	A2	A2	A2	A2	A2	A2
P1 O1 C1	P2 O2 C2	P3 O3 C3	P4 O4 C4	P5 O5 C5	P6 O6 C6	P7 O7 C7	P8 O8 C8

30 / 2L2-1

/ 2M2-1

PANEL SECTION

FIELD SECTION

X-DO2 2A

17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
		2A		2A		2A		2A		2A		2A		2A	

TAG :	ABV-07 CMD	ABV-08 CMD	ABV-09 CMD	ABV-10 CMD	ABV-11 CMD	ABV-12 CMD	ABV-13 CMD	ABV-14 CMD
DESC :	CIP LINE DRAIN VALVE OPEN/CLOSE CMD	WINCH CIP INLET VALVE OPEN/CLOSE CMD	SCRUBBER TANK DRAIN VALVE OPEN/CLOSE CMD	SCRUBBER TANK WATER INLET VALVE OPEN/CLOSE CMD	CYCLONE CIP OUTLET VALVE OPEN/CLOSE CMD	CHAMBER CIP OUTLET VALVE OPEN/CLOSE CMD	CIP TANK RETURN CIP INLET VALVE OPEN/CLOSE CMD	FEED TANK CIP INLET VALVE OPEN/CLOSE CMD

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 2A

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025	DRAWN BY	S.B.
02	AS PER COMMENTS	21-01-2025	CHECKED	P.B.
01	AS PER COMMENTS	14-01-2025		
00	FOR APPROVAL	25-11-2024		

TITLE:- 16DO-DIGITAL OUTPUT WIRING-2
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697
DRG. NO.:- 2024/PRG/MSD/697
SHEET NO. 32 / 43
REV.NO. AS BUILT

DO3 | 16 DO DIGITAL OUTPUT MODULE-3 | 6ES71326BH010BA0 | DQ 16x24 VDC | 1 OF 2 | SIEMENS

CH0 CH1 CH2 CH3 CH4 CH5 CH6 CH7

Q4.0 1 | Q4.1 2 | Q4.2 3 | Q4.3 4 | Q4.4 5 | Q4.5 6 | Q4.6 7 | Q4.7 8

Q4:0/0 | Q4:0/1 | Q4:0/2 | Q4:0/3 | Q4:0/4 | Q4:0/5 | Q4:0/6 | Q4:0/7

11.7 / 2M2-7

RELAY BOARD-5
8 CH (RB-05) 24VDC-1C/0

-1 +1	-2 +2	-3 +3	-4 +4	-5 +5	-6 +6	-7 +7	-8 +8
A1	A1	A1	A1	A1	A1	A1	A1
RB05-R1	RB05-R2	RB05-R3	RB05-R4	RB05-R5	RB05-R6	RB05-R7	RB05-R8
A2	A2	A2	A2	A2	A2	A2	A2
P1 O1 C1	P2 O2 C2	P3 O3 C3	P4 O4 C4	P5 O5 C5	P6 O6 C6	P7 O7 C7	P8 O8 C8

11.1 / 2L2-2 → 2L2-2 / 32

11.3 / 2M2-2 → 2M2-2 / 32

PANEL SECTION

X-DO3 2A: 1 2 | 2A 3 4 | 2A 5 6 | 2A 7 8 | 2A 9 10 | 2A 11 12 | 2A 13 14 | 2A 15 16

FIELD SECTION

TAG :	DESC :
ABV-15 CMD	CIP VALVE-01 OPEN/CLOSE CMD
ABV-16 CMD	CIP TANK FRESH WATER INLET VALVE OPEN/CLOSE CMD
ADV-01 CMD	FEED TANK FEED INLET VALVE OPEN/CLOSE CMD
ADV-02 CMD	FEED TANK FEED OUTLET VALVE OPEN/CLOSE CMD
ADV-03 CMD	WATER TANK OUTLET VALVE OPEN/CLOSE CMD
AGV-01 CMD	CIP TANK STEAM INLET VALVE OPEN/CLOSE CMD
SOV-01 CMD	CHAMBER CIP NOZZLE SOV
SOV-02 CMD	POWDER DUCTCIP NOZZLE SOV

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 2A

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025	DRAWN BY	S.B.
02	AS PER COMMENTS	21-01-2025	CHECKED	P.B.
01	AS PER COMMENTS	14-01-2025		
00	FOR APPROVAL	25-11-2024		

TITLE:- 16DO-DIGITAL OUTPUT WIRING-3
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697
DRG. NO.:- 2024/PRG/MSD/697
SHEET NO. 33 / 43
REV.NO. AS BUILT

DO3 | 16 DO DIGITAL OUTPUT MODULE-3 | 6ES71326BH010BA0 | DQ 16x24 VDC | 2 OF 2 | SIEMENS

CH8 CH9 CH10 CH11 CH12 CH13 CH14 CH15

Q5.0 1 | Q5.1 2 | Q5.2 3 | Q5.3 4 | Q5.4 5 | Q5.5 6 | Q5.6 7 | Q5.7 8

Q5:0/0 | Q5:0/1 | Q5:0/2 | Q5:0/3 | Q5:0/04 | Q5:0/5 | Q5:0/6 | Q5:0/7

11.7 / 2M2-8

RELAY BOARD-6
8 CH (RB-06) 24VDC-1C/0

-1 +1	-2 +2	-3 +3	-4 +4	-5 +5	-6 +6	-7 +7	-8 +8
A1	A1	A1	A1	A1	A1	A1	A1
RB06-R1	RB06-R2	RB06-R3	RB06-R4	RB06-R5	RB06-R6	RB06-R7	RB06-R8
A2	A2	A2	A2	A2	A2	A2	A2
P1 O1 C1	P2 O2 C2	P3 O3 C3	P4 O4 C4	P5 O5 C5	P6 O6 C6	P7 O7 C7	P8 O8 C8

32 / 2L2-2

32 / 2M2-2

PANEL SECTION

FIELD SECTION

X-DO3 2A: 17 18 | 2A 19 20 | 2A 21 22 | 2A 23 24 | 2A 25 26 | 2A 27 28 | 2A 29 30

x1 HOOTER x2

TAG :	DESC :
SOV-03 CMD	CY-01 CIP NOZZLE SOV
PH-01 CMD	PNEMATIC HAMMER -01 SS CMD
PH-02 CMD	PNEMATIC HAMMER -02 SS CMD
PH-03 CMD	PNEMATIC HAMMER -03 SS CMD
PH-04 CMD	PNEMATIC HAMMER -04 SS CMD
PH-05 CMD	PNEMATIC HAMMER -05 SS CMD
PH-06	PNEUMATIC HAMMER PH-06 CMD
-	HOOTER

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 2A

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE		
03	AS BUILT	28-02-2025	TITLE:- 16DO-DIGITAL OUTPUT WIRING-3	
02	AS PER COMMENTS	21-01-2025	FILE NO:	MES / E&I / MSD-697
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.
			APPROVED	MESL

PROJECT:- SPRAY DRYER

PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697 | SHEET NO. 34 / 43 | REV.NO. AS BUILT

DO4 16 DO DIGITAL OUTPUT MODULE-4 6ES71326BH010BA0 DQ 16x24 VDC 1 OF 2 SIEMENS

CH0 CH1 CH2 CH3 CH4 CH5 CH6 CH7

Q6.0 1 Q6.1 2 Q6.2 3 Q6.3 4 Q6.4 5 Q6.5 6 Q6.6 7 Q6.7 8

Q6:0/0 Q6:0/1 Q6:0/2 Q6:0/3 Q6:0/4 Q6:0/5 Q6:0/6 Q6:0/7

11.8 / 2M2-9

RELAY BOARD-7
8 CH (RB-07) 24VDC-1C/0

-1 +1 A1 RB07-R1 A2 | -2 +2 A1 RB07-R2 A2 | -3 +3 A1 RB07-R3 A2 | -4 +4 A1 RB07-R4 A2 | -5 +5 A1 RB07-R5 A2 | -6 +6 A1 RB07-R6 A2 | -7 +7 A1 RB07-R7 A2 | -8 +8 A1 RB07-R8 A2

P1 C1 O1 | P2 C2 O2 | P3 C3 O3 | P4 C4 O4 | P5 C5 O5 | P6 C6 O6 | P7 C7 O7 | P8 C8 O8

PANEL SECTION

X-DO4 2A 1 | 2 | 2A 3 | 4 | 2A 5 | 6 | 2A 7 | 8 | 2A 9 | 10 | 2A 11 | 12 | 2A 13 | 14 | 2A 15 | 16

FIELD SECTION

TAG :	SP-01	SP-02	SP-03	SP-04	SP-05	SP-06	SP-07	SP-08
DESC :	SPARE-01	SPARE-02	SPARE-03	SPARE-04	SPARE-05	SPARE-06	SPARE-07	SPARE-08

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 2A

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025	DRAWN BY	S.B.
02	AS PER COMMENTS	21-01-2025	CHECKED	P.B.
01	AS PER COMMENTS	14-01-2025		
00	FOR APPROVAL	25-11-2024		

TITLE:- 16DO-DIGITAL OUTPUT WIRING-4
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697
DRG. NO.:- 2024/PRG/MSD/697
SHEET NO. 35 / 43
REV.NO. AS BUILT

DO4 | 16 DO DIGITAL OUTPUT MODULE-4 | 6ES71326BH010BA0 | DQ 16x24 VDC | 2 OF 2 | SIEMENS

CH8 CH9 CH10 CH11 CH12 CH13 CH14 CH15

Q7.0 1 | Q7.1 2 | Q7.2 3 | Q7.3 4 | Q7.4 5 | Q7.5 6 | Q7.6 7 | Q7.7 8

Q7:0/0 | Q7:0/1 | Q7:0/2 | Q7:0/3 | Q7:0/4 | Q7:0/5 | Q7:0/6 | Q7:0/7

11.9 / 2M2-10

RELAY BOARD-8
8 CH (RB-08) 24VDC-1C/0

-1 +1	-2 +2	-3 +3	-4 +4	-5 +5	-6 +6	-7 +7	-8 +8
A1	A1	A1	A1	A1	A1	A1	A1
RB08-R1	RB08-R2	RB08-R3	RB08-R4	RB08-R5	RB08-R6	RB08-R7	RB08-R8
A2	A2	A2	A2	A2	A2	A2	A2
P1 C1 O1	P2 C2 O2	P3 C3 O3	P4 C4 O4	P5 C5 O5	P6 C6 O6	P7 C7 O7	P8 C8 O8

PANEL SECTION

X-DO4 2A 17 | 18 | 2A 19 | 20 | 2A 21 | 22 | 2A 23 | 24 | 2A 25 | 26 | 2A 27 | 28 | 2A 29 | 30 | 2A 31 | 32

FIELD SECTION

TAG :	SP-09	SP-10	SP-11	SP-12	SP-13	SP-14	SP-15	SP-16
DESC :	SPARE-09	SPARE-10	SPARE-11	SPARE-12	SPARE-13	SPARE-14	SPARE-15	SPARE-16

NOTE-TYPE OF TERMINALS USED
1 -NORMAL TERMINAL
2 -FUSED TERMINAL 2A

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REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 16DO-DIGITAL OUTPUT WIRING-4
FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER
PROJECT:- MSD-697
DRG. NO.:- 2024/PRG/MSD/697
SHEET NO. 36 / 43
REV.NO. AS BUILT

AI1 | 8 AI ANALOG INPUT MODULE-1 | 6ES71346GF000AA1 | AI 8xI 2-/4-WIRE | 1 OF 1 | SIEMENS

CH0 CH1 CH2 CH3 CH4 CH5 CH6 CH7

L+ M I0+ 1 UV 9 I1+ 2 UV 10 I2+ 3 UV 11 I3+ 4 UV 12 I4+ 5 UV 13 I5+ 6 UV 14 I6+ 7 UV 15 I7+ 8 UV 16

9.3 / 3L1

9.3 / 3M1

0.5mm²BR 0.5mm²OG

AI1-UV1 PIW300+ AI1-UV2 PIW302+ AI1-UV3 PIW304+ AI1-UV4 PIW306+ AI1-UV5 PIW308+ AI1-UV6 PIW310+ AI1-UV7 PIW312+ AI1-UV8 PIW314+

IE

X-AI1 250mA 1 2 250mA 3 4 250mA 5 6 250mA 7 8 250mA 9 10 250mA 11 12 250mA 13 14 250mA 15 16

PANEL SECTION

FIELD SECTION

TAG :	DESC :	
TT-101	CIP TANK TEMP. TRANSMITTER	(2 WIRE)
TT-102	SPRAY DRYER INLET TEMP. TRANSMITTER-1	(2 WIRE)
TT-103	SPRAY DRYER INLET TEMP. TRANSMITTER-2	(2 WIRE)
TT-104	SPRAY DRYER OUTLET TEMP. TRANSMITTER	(2 WIRE)
TT-105	IFBD INLET TEMP. TRANSMITTER	(2 WIRE)
PT-101	SPRAY DRYER CHAMBER PRESSURE (SD-01)	(2 WIRE)
PT-102	FEED LINE PRESSURE	(2 WIRE)
DPT-101	PRESSURE BETWEEN CHAMBER & IFBD	(2 WIRE)

NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL 250mA

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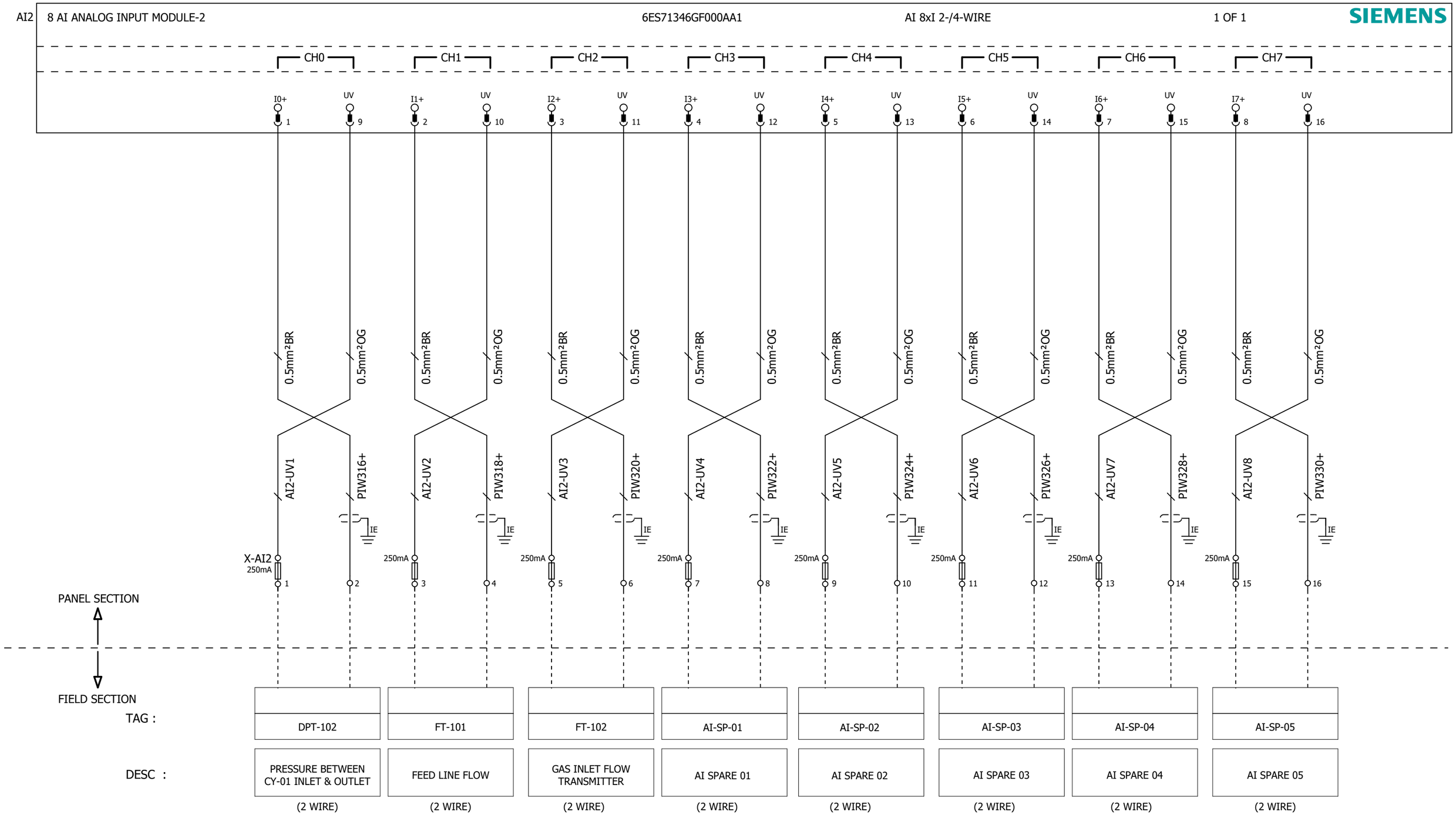
REVISION	DETAILS	DATE			
03	AS BUILT	28-02-2025	TITLE:- 8AI-ANALOG INPUT WIRING-1		
02	AS PER COMMENTS	21-01-2025	FILE NO:	MES / E&I / MSD-697	
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.	
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.	
			APPROVED	MESL	

PROJECT:- SPRAY DRYER

PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697 | SHEET NO. 37 / 43 | REV.NO. AS BUILT

AI2 | 8 AI ANALOG INPUT MODULE-2 | 6ES71346GF000AA1 | AI 8xI 2-/4-WIRE | 1 OF 1 | SIEMENS



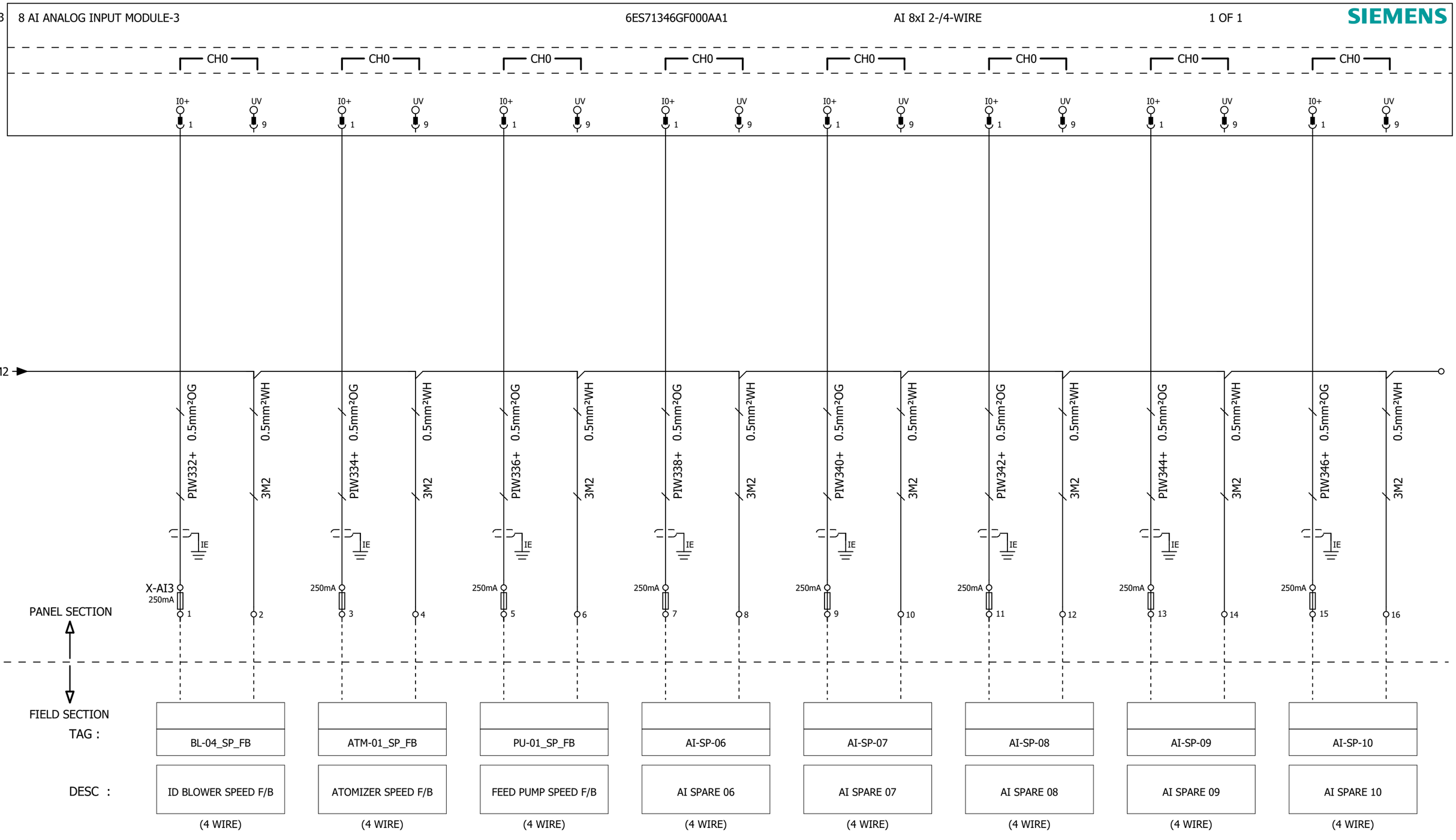
NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL 250mA

MOJJ ENGINEERING SYSTEMS LTD	03	AS BUILT	28-02-2025	TITLE:-8AI-ANALOG INPUT WIRING-2	PROJECT:- SPRAY DRYER
	02	AS PER COMMENTS	21-01-2025	FILE NO: MES / E&I / MSD-697	
	01	AS PER COMMENTS	14-01-2025	DRAWN BY S.B.	PROJECT:- MSD-697
	00	FOR APPROVAL	25-11-2024	CHECKED P.B.	DRG. NO.:- 2024/PRG/MSD/697 / SHEET NO. 38 / 43 / REV.NO. AS BUILT
	REVISION	DETAILS	DATE	APPROVED MESL	

AI3 | 8 AI ANALOG INPUT MODULE-3 | 6ES71346GF000AA1 | AI 8xI 2-/4-WIRE | 1 OF 1 | SIEMENS



NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL 250mA

03	AS BUILT	28-02-2025	TITLE:- 8AI-ANALOG INPUT WIRING-3		PROJECT:- SPRAY DRYER
02	AS PER COMMENTS	21-01-2025	FILE NO: MES / E&I / MSD-697		PROJECT:- MSD-697
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.	DRG. NO.:- 2024/PRG/MSD/697
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.	SHEET NO. 39 / 43
REVISION	DETAILS	DATE	APPROVED	MESL	REV.NO. AS BUILT

MOJJ ENGINEERING SYSTEMS LTD

AO1 4 AO ANALOG OUTPUT MODULE-1 | 6ES71356HD000BA1 | AO 4xI 2-/4-WIRE | 1 OF 1 | SIEMENS

CH0 CH1 CH2 CH3

L+ 1 | M 5 | Q0+ 1 | Q0- 5 | Q1+ 2 | Q1- 6 | Q2+ 3 | Q2- 7 | Q3+ 4 | Q3- 8

9.4 / 4L1

9.4 / 4M1

0.5mm²OG PQW300+ | 0.5mm²WH PQW300- | 0.5mm²OG PQW302+ | 0.5mm²WH PQW302- | 0.5mm²OG PQW304+ | 0.5mm²WH PQW304- | 0.5mm²OG PQW306+ | 0.5mm²WH PQW306-

X-AO1 250mA 1 | 2 | 250mA 3 | 4 | 250mA 5 | 6 | 250mA 7 | 8

PANEL SECTION

FIELD SECTION

TAG :	AGV-02	EH-01	BL-04_SP	ATM-01_SP
DESC :	CIP TANK TEMP. CONTROL VALVE	ELECTRIC HEATER REMOTE SET POINT	ID BLOWER SPEED REFERENCE	ATOMIZER SPEED REFERENCE
RANGE :	-	-	-	-

NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL 250mA

MOJJ ENGINEERING SYSTEMS LTD

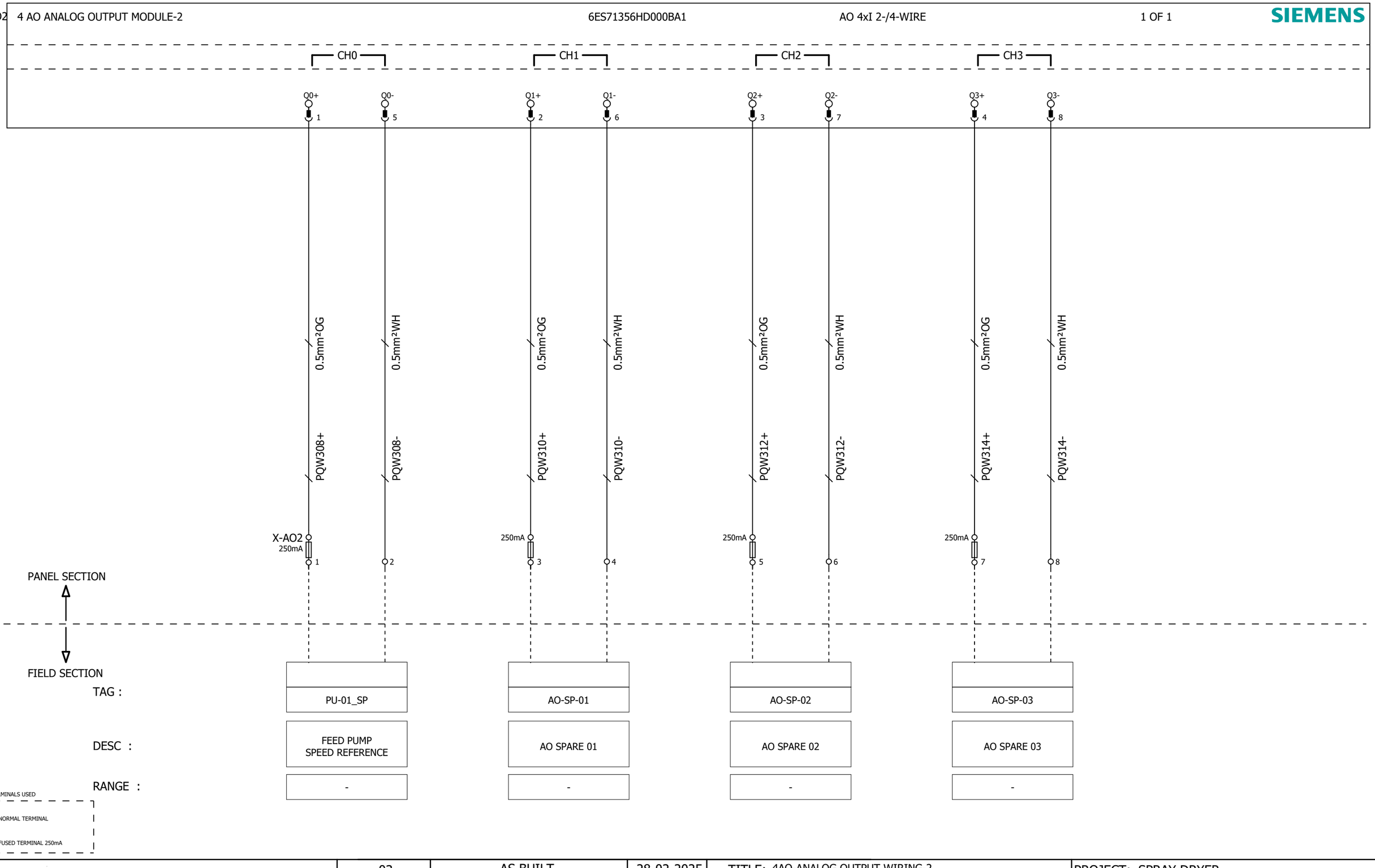
REVISION	DETAILS	DATE		
03	AS BUILT	28-02-2025	TITLE:-4AO-ANALOG OUTPUT WIRING-1	
02	AS PER COMMENTS	21-01-2025	FILE NO: MES / E&I / MSD-697	
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.
			APPROVED	MESL

PROJECT:- SPRAY DRYER

PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697 | SHEET NO. 40 / 43 | REV.NO. AS BUILT

AO2 4 AO ANALOG OUTPUT MODULE-2 | 6ES71356HD000BA1 | AO 4xI 2-/4-WIRE | 1 OF 1 | SIEMENS



NOTE-TYPE OF TERMINALS USED

1 -NORMAL TERMINAL

2 -FUSED TERMINAL 250mA

MOJJ ENGINEERING SYSTEMS LTD

REVISION	DETAILS	DATE	APPROVED	MESL
03	AS BUILT	28-02-2025		
02	AS PER COMMENTS	21-01-2025		
01	AS PER COMMENTS	14-01-2025	DRAWN BY	S.B.
00	FOR APPROVAL	25-11-2024	CHECKED	P.B.

TITLE:- 4AO-ANALOG OUTPUT WIRING-2

FILE NO: MES / E&I / MSD-697

PROJECT:- SPRAY DRYER

PROJECT:- MSD-697

DRG. NO.:- 2024/PRG/MSD/697 | SHEET NO. 41 / 43 | REV.NO. AS BUILT

BILL OF MATERIAL

MCC SECTION

SR.NO.	ITEM DESCRIPTION	MAKE	PART NO	QTY.
1	MAIN SWITCH 3P, 32AMP	SALZER	-	1
2	32A, 2P, 10kA, C-Curve MCB	SIEMENS	5SL42327RC	1
3	6A, 2P, 10kA, C-Curve MCB	SIEMENS	5SL42067RC	5
4	10A, 2P, 10kA, C-Curve MCB	SIEMENS	5SL42107RC	2
5	6A, 1P, 10kA, C-Curve MCB	SIEMENS	5SL41067RC	9
6	20A, 2P, 10kA, C-Curve MCB	SIEMENS	5SL42207RC	3
7	40A, 2P, 10kA, C-Curve MCB	SIEMENS	5SL42407RC	1
8	Fuse Terminals with 2Amp Fuse	ELMEX	-	20
9	230VAC On Indication (RED)	Teknic	3PLBR4L	1
10	24VDC On Indication (GREEN)	Teknic	3PLBR3L	4
11	24VDC, 10AMP SMPS Module	SIEMENS	-	2
12	24VDC, 20AMP Redundancy Module	Weidmuller	-	1
13	24VDC, 20AMP SMPS Module	SIEMENS	-	2
14	24VDC, 40AMP Redundancy Module	Weidmuller	-	1
15	4Sq.mm Control Terminals (Blue)	Teknic	-	10
16	4Sq.mm Control Terminals (Grey)	Teknic	-	10
17	10Sq.mm Power Terminal (Red)	Teknic	-	1
18	10Sq.mm Power Terminal (Black)	Teknic	-	1
19	Door Limit Switch	Kaycee	-	2
20	5 Watt, Tubelight 230VAC supply voltage	Oreva	-	2
21	6 Inch Air Filter	Elttro	-	5
22	6 Inch Fan	Rexnord	-	2
23	Reset Push Button (Yellow)	Teknic	-	1
24	Hooter Acknowledge Push Button (Black)	Teknic	-	1
25	Emergency Push Button	Teknic	-	1
26	Hooter	Shavison	-	1

BILL OF MATERIAL

PLC SECTION

SR.NO.	ITEM DESCRIPTION	MAKE	PART NO	QTY.
1	CPU-1512SP-1 PN FOR ET 200 SP	SIEMENS	6ES75121DM030AB0	1
2	PROFIBUS CARD	SIEMENS	6ES75455DA000AB0	1
3	SIMATIC ET 200SP, Digital input module, DI 16x 24V DC	SIEMENS	6ES71316BH010BA0	7
4	SIMATIC ET 200SP, Digital output module, DQ 16x 24V	SIEMENS	6ES71326BH010BA0	4
5	SIMATIC ET 200SP, analog input module, AI 8XU/I 2/4-wire basic	SIEMENS	6ES71346GF000AA1	3
6	SIMATIC ET 200SP, Analog output module, AQ 4XU/I	SIEMENS	6ES71356HD000BA1	2
7	Unmanaged Industrial Ethernet Switch	SIEMENS	-	1
8	SIMATIC ET 200SP, BaseUnit BU15-P16+A0+2D, BU type A0, push-in terminals	SIEMENS	6ES71936BP000DA0	4
9	SIMATIC ET 200SP, BaseUnit BU15-P16+A0+2B, BU type A0, Push-in terminals	SIEMENS	6ES71936BP000BA0	12
10	SIMATIC S7 MEMORY CARD 12 MB FOR PLC	SIEMENS	6ES7954-8LE03-0AA0	1
11	SIMATIC S7 MEMORY CARD 2GB FOR HMI	SIEMENS	6AV21818XP000AX0	1
12	SIMATIC HMI MTP1200 UNIFIED COMFORT	SIEMENS	6AV21283MB060AX0	1
13	2.5 Sq.mm Control Terminals	Elmex	-	220
14	Fuse Terminals	Elmex	-	190
15	250mA Fuse	Std.	-	32
16	500mA Fuse	Std.	-	95
17	2A Fuse	Std.	-	64
18	8 Channel 24VDC Relay Board	Omron	-	8